Linear Algebra and Its Applications

Chapter 8

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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8.1 Affine combinations

An affine combination is a linear combination whose coefficients sum to 1. They may be negative: nonnegativity is an additional condition for a *convex* combination, studied later. The affine hull aff S is the set of all affine combinations from S.

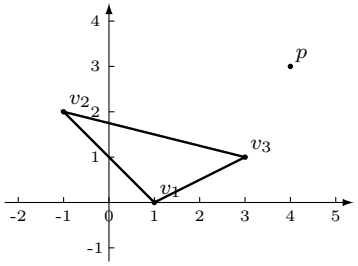
For nonempty $S = \{v_1, \ldots, v_k\}$,

$$\operatorname{aff} S = v_1 + \operatorname{Span}\{v_2 - v_1, \ldots, v_k - v_1\}.$$

An equivalent computational method appends a coordinate 1 to each point. Solving $[\widetilde{v}_1 \ \cdots \ \widetilde{v}_k]c = \widetilde{p}$ then enforces both the point equation and $\sum c_j = 1$. An affine set is closed under affine combinations; it may be empty under the usual closure convention. A nonempty affine set is a translated subspace, or flat.

Practice problem 8.1.P1: Three noncollinear points fill the plane affinely

The edge differences from v_1 are $v_2 - v_1 = (-2, 2)$ and $v_3 - v_1 = (2, 1)$. Their determinant is $(-2)1 - 2 \cdot 2 = -6 \neq 0$, so they span $\mathbb{R}^2$. Hence the affine hull of the three points is all of $\mathbb{R}^2$, and it must contain p.



For explicit weights solve $p - v_1 = c_2(v_2 - v_1) + c_3(v_3 - v_1)$:

$$-2c_2 + 2c_3 = 3, \qquad 2c_2 + c_3 = 3.$$

Adding suitable multiples gives $c_2 = 1/2$, $c_3 = 2$, and then $c_1 = 1 - c_2 - c_3 = -3/2$. Thus $p = -\frac{3}{2}v_1 + \frac{1}{2}v_2 + 2v_3$. These weights sum to 1 and reproduce $(4, 3)$; a negative weight is permitted for an affine combination.

Exercise 8.1.1: Require the weights to sum to one

To require both the point equation and the coefficient-sum condition, solve

$$Hc = \widetilde{y}, \qquad H = \begin{bmatrix} 1 & -2 & 0 & 3 \\ 2 & 2 & 4 & 7 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \quad \widetilde{y} = \begin{bmatrix} 5 \\ 3 \\ 1 \end{bmatrix}.$$

The last row of H enforces $c_1 + \cdots + c_k = 1$. Row reduction of the augmented system gives

$$\operatorname{rref}[H \mid \widetilde{y}] = \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & -\frac{3}{2} & -\frac{3}{2} \\ 0 & 0 & 1 & \frac{5}{2} & \frac{1}{2} \end{bmatrix}.$$

There is a free parameter; setting it to zero gives one valid choice. Other choices are also allowed, since an affine representation need not be unique for a dependent set. One affine coefficient column is

$$c = \begin{bmatrix} 2 \\ -\frac{3}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix}.$$

Its entries sum to 1, and substituting them in $\sum c_j v_j$ gives exactly $\begin{bmatrix} 5 \\ 3 \end{bmatrix}$. Thus the requested affine combination exists.

Exercise 8.1.2: Require the weights to sum to one

To require both the point equation and the coefficient-sum condition, solve

$$Hc = \widetilde{y}, \qquad H = \begin{bmatrix} 1 & -1 & 3 \\ 1 & 2 & 2 \\ 1 & 1 & 1 \end{bmatrix}, \quad \widetilde{y} = \begin{bmatrix} 5 \\ 7 \\ 1 \end{bmatrix}.$$

The last row of H enforces $c_1 + \cdots + c_k = 1$. Row reduction of the augmented system gives

$$\operatorname{rref}[H \mid \widetilde{y}] = \begin{bmatrix} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{bmatrix}.$$

One affine coefficient column is

$$c = \begin{bmatrix} -5 \\ 2 \\ 4 \end{bmatrix}.$$

Its entries sum to 1, and substituting them in $\sum c_j v_j$ gives exactly $\begin{bmatrix} 5 \\ 7 \end{bmatrix}$. Thus the requested affine combination exists.

Exercise 8.1.3: Require the weights to sum to one

To require both the point equation and the coefficient-sum condition, solve

$$Hc = \widetilde{y}, \qquad H = \begin{bmatrix} -3 & 0 & 4 \\ 1 & 4 & -2 \\ 1 & -2 & 6 \\ 1 & 1 & 1 \end{bmatrix}, \quad \widetilde{y} = \begin{bmatrix} 17 \\ 1 \\ 5 \\ 1 \end{bmatrix}.$$

The last row of H enforces $c_1 + \cdots + c_k = 1$. Row reduction of the augmented system gives

$$\operatorname{rref}[H \mid \widetilde{y}] = \begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

One affine coefficient column is

$$c = \begin{bmatrix} -3 \\ 2 \\ 2 \end{bmatrix}.$$

Its entries sum to 1, and substituting them in $\sum c_j v_j$ gives exactly $\begin{bmatrix} 17 \\ 1 \\ 5 \end{bmatrix}$. Thus the requested affine combination exists.

Exercise 8.1.4: Require the weights to sum to one

To require both the point equation and the coefficient-sum condition, solve

$$Hc = \widetilde{y}, \qquad H = \begin{bmatrix} 1 & 2 & 4 \\ 2 & -6 & 3 \\ 0 & 7 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad \widetilde{y} = \begin{bmatrix} -3 \\ 4 \\ -4 \\ 1 \end{bmatrix}.$$

The last row of H enforces $c_1 + \cdots + c_k = 1$. Row reduction of the augmented system gives

$$\operatorname{rref}[H \mid \widetilde{y}] = \begin{bmatrix} 1 & 0 & 0 & \frac{13}{5} \\ 0 & 1 & 0 & -\frac{2}{5} \\ 0 & 0 & 1 & -\frac{6}{5} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

One affine coefficient column is

$$c = \begin{bmatrix} \frac{13}{5} \\ -\frac{2}{5} \\ -\frac{6}{5} \end{bmatrix}.$$

Its entries sum to 1, and substituting them in $\sum c_j v_j$ gives exactly $\begin{bmatrix} -3 \\ 4 \\ -4 \end{bmatrix}$. Thus the requested affine combination exists.

Exercise 8.1.5: Use orthogonality before checking the affine condition

The given orthogonal basis has squared lengths $6, 5, 30$. Orthogonal expansion therefore gives the unique linear coefficients

$$c_1 = \frac{p \cdot b_1}{6}, \qquad c_2 = \frac{p \cdot b_2}{5}, \qquad c_3 = \frac{p \cdot b_3}{30}.$$

Because the basis is linearly independent, these are the only possible coefficients. The representation is affine exactly when their sum is 1.

(a) The point $p = \begin{bmatrix} 3 \\ 8 \\ 4 \end{bmatrix}$.

The dot products with b_1, b_2, b_3 are $18,\ -5,\ -30$, respectively. Dividing by $6, 5, 30$ gives

$$c = \begin{bmatrix} 3 \\ -1 \\ -1 \end{bmatrix}, \qquad c_1 + c_2 + c_3 = 1.$$

Thus $p = (3)b_1 + (-1)b_2 + (-1)b_3$ is an affine combination.

(b) The point $p = \begin{bmatrix} 6 \\ -3 \\ 3 \end{bmatrix}$**.**

The dot products with b_1, b_2, b_3 are $12,\ 0,\ 30$, respectively. Dividing by $6, 5, 30$ gives

$$c = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \qquad c_1 + c_2 + c_3 = 3.$$

The unique linear coefficients do not sum to 1, so no affine combination of these basis points equals p.

(c) The point $p = \begin{bmatrix} 0 \\ -1 \\ -5 \end{bmatrix}$**.**

The dot products with b_1, b_2, b_3 are $-6,\ 10,\ 0$, respectively. Dividing by $6, 5, 30$ gives

$$c = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, \qquad c_1 + c_2 + c_3 = 1.$$

Thus $p = (-1)b_1 + (2)b_2 + (0)b_3$ is an affine combination.

Exercise 8.1.6: Use orthogonality before checking the affine condition

The given orthogonal basis has squared lengths $6, 5, 30$. Orthogonal expansion therefore gives the unique linear coefficients

$$c_1 = \frac{p \cdot b_1}{6}, \qquad c_2 = \frac{p \cdot b_2}{5}, \qquad c_3 = \frac{p \cdot b_3}{30}.$$

Because the basis is linearly independent, these are the only possible coefficients. The representation is affine exactly when their sum is 1.

(a) The point $p = \begin{bmatrix} 0 \\ -19 \\ -5 \end{bmatrix}$**.**

The dot products with b_1, b_2, b_3 are $-24,\ 10,\ 90$, respectively. Dividing by $6, 5, 30$ gives

$$c = \begin{bmatrix} -4 \\ 2 \\ 3 \end{bmatrix}, \qquad c_1 + c_2 + c_3 = 1.$$

Thus $p = (-4)b_1 + (2)b_2 + (3)b_3$ is an affine combination.

(b) The point $p = \begin{bmatrix} \frac{3}{2} \\ -\frac{13}{10} \\ -\frac{1}{2} \end{bmatrix}$**.**

The dot products with b_1, b_2, b_3 are $\frac{6}{5},\ \frac{5}{2},\ 9$, respectively. Dividing by $6, 5, 30$ gives

$$c = \begin{bmatrix} \frac{1}{5} \\ \frac{1}{2} \\ \frac{3}{10} \end{bmatrix}, \qquad c_1 + c_2 + c_3 = 1.$$

Thus $p = (\frac{1}{5})b_1 + (\frac{1}{2})b_2 + (\frac{3}{10})b_3$ is an affine combination.

(c) The point $p = \begin{bmatrix} 5 \\ -4 \\ 0 \end{bmatrix}$**.**

The dot products with b_1, b_2, b_3 are $6,\ 5,\ 30$, respectively. Dividing by $6, 5, 30$ gives

$$c = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \qquad c_1 + c_2 + c_3 = 3.$$

The unique linear coefficients do not sum to 1, so no affine combination of these basis points equals p.

Exercise 8.1.7: Distinguish a linear span from an affine hull

Place the three given independent vectors in the columns of

$$V = \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 2 \\ 3 & 0 & 1 \\ 0 & 4 & 1 \end{bmatrix}.$$

A target is in $\operatorname{Span} S$ if $Vc = p$ is consistent; it is in $\operatorname{aff} S$ only if its coefficients also sum to 1. Independence makes a consistent coefficient column unique.

(a) Test p_1**.**

$$\operatorname{rref}[V \mid p_1] = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The system is consistent with $c = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$, so the point is in $\operatorname{Span} S$. The sum is 2, so the point is not in $\operatorname{aff} S$.

(b) Test p_2**.**

$$\operatorname{rref}[V \mid p_2] = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The system is consistent with $c = \begin{bmatrix} 2 \\ -4 \\ 3 \end{bmatrix}$, so the point is in $\operatorname{Span} S$. The sum is 1, so the point is also in $\operatorname{aff} S$.

(c) Test p_3.

$$\operatorname{rref}[V \mid p_3] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The augmented last column has a pivot, so the system is inconsistent. The point is not in $\operatorname{Span} S$ and therefore cannot be in the smaller set $\operatorname{aff} S$.

Exercise 8.1.8: Distinguish a linear span from an affine hull

Place the three given independent vectors in the columns of

$$V = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 3 & 6 & 12 \\ -2 & -5 & -6 \end{bmatrix}.$$

A target is in $\operatorname{Span} S$ if $Vc = p$ is consistent; it is in $\operatorname{aff} S$ only if its coefficients also sum to 1. Independence makes a consistent coefficient column unique.

(a) Test p_1.

$$\operatorname{rref}[V \mid p_1] = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The system is consistent with $c = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$, so the point is in $\operatorname{Span} S$. The sum is 3, so the point is not in $\operatorname{aff} S$.

(b) Test p_2.

$$\operatorname{rref}[V \mid p_2] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The augmented last column has a pivot, so the system is inconsistent. The point is not in Span S and therefore cannot be in the smaller set aff S.

(c) Test p_3.

$$\operatorname{rref}[V \mid p_3] = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The system is consistent with $c = \begin{bmatrix} -2 \\ 6 \\ -3 \end{bmatrix}$, so the point is in Span S. The sum is 1, so the point is also in aff S.

Exercise 8.1.9: Represent a translated line by two points

Write the free scalar as t to avoid confusing it with a vector coordinate. Choose

$$v_1 = p = \begin{bmatrix} -3 \\ 0 \end{bmatrix}, \qquad v_2 = p + u = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Then $p + tu = (1 - t)v_1 + tv_2$, whose two weights sum to 1. Conversely any affine combination of v_1, v_2 has precisely this form. Thus the entire solution set is aff$\{v_1, v_2\}$, as required.

Exercise 8.1.10: Represent a translated line by two points

Write the free scalar as t to avoid confusing it with a vector coordinate. Choose

$$v_1 = p = \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix}, \qquad v_2 = p + u = \begin{bmatrix} 6 \\ -2 \\ 2 \end{bmatrix}.$$

Then $p + tu = (1 - t)v_1 + tv_2$, whose two weights sum to 1. Conversely any affine combination of v_1, v_2 has precisely this form. Thus the entire solution set is aff$\{v_1, v_2\}$, as required.

Exercise 8.1.11: Affine closure, flats, and hull identities

(a) **True.** This is the definition of the affine hull.

(b) **False.** For $S = \{e_1, e_2\} \subset \mathbb{R}^2$, zero is a linear combination but not an affine combination: its unique linear coefficients are $(0, 0)$, whose sum is 0, not 1.

(c) **True.** The hull is $v_1 + \operatorname{Span}\{v_2 - v_1\}$, a one-dimensional flat when $v_2 \neq v_1$.

(d) **False.** A flat is a translated subspace and need not contain zero. The line $\{(x, 1) : x \in \mathbb{R}\}$ is a flat but not a subspace.

(e) **True.** A plane in $\mathbb{R}^3$ has dimension $2 = 3 - 1$, which is exactly the hyperplane dimension.

Exercise 8.1.12: Affine closure, flats, and hull identities

(a) **False.** The only affine combinations of one point x use total coefficient 1, so $\operatorname{aff}\{x\} = \{x\}$.

(b) **True.** If a set contains its affine hull, it contains every affine combination of its points, which is the closure condition defining an affine set. Conversely an affine set contains all those combinations. Every set is contained in its own affine hull, so containment here is actually equality.

(c) **True.** A one-dimensional flat is a line.

(d) **False in general.** A hyperplane in $\mathbb{R}^n$ has dimension $n-1$. A two-dimensional flat is a hyperplane in $\mathbb{R}^3$, but not in $\mathbb{R}^4$.

(e) **True.** Write the flat as $p + W$ with W a subspace. If $0 = p + w$ for some $w \in W$, then $p = -w \in W$, so $p + W = W$ is a subspace.

Exercise 8.1.13: Affine closure, flats, and hull identities

Suppose $c_2(v_2 - v_1) + c_3(v_3 - v_1) = 0$. Regrouping gives $(-c_2 - c_3)v_1 + c_2 v_2 + c_3 v_3 = 0$. Since v_1, v_2, v_3 form a basis, all three coefficients are zero, in particular $c_2 = c_3 = 0$. Thus the two differences are independent. Their span therefore has dimension 2 and is a plane through zero in $\mathbb{R}^3$.

Exercise 8.1.14: Affine closure, flats, and hull identities

The affine-hull identity gives

$$\operatorname{aff}\{v_1, v_2, v_3\} = v_1 + \operatorname{Span}\{v_2 - v_1, v_3 - v_1\}.$$

By Exercise 13 the difference span is two-dimensional, so this is a plane. It contains v_1, $v_2 = v_1 + (v_2 - v_1)$, and $v_3 = v_1 + (v_3 - v_1)$. Conversely any plane containing those three points must contain every affine combination of them, so this is precisely the plane through them. The independence of the differences rules out a line instead.

Exercise 8.1.15: Affine closure, flats, and hull identities

Take solutions $x_1, \ldots, x_k$ and coefficients c_j with $\sum_j c_j = 1$. Matrix linearity gives

$$A\left(\sum_j c_j x_j\right) = \sum_j c_j A x_j = \sum_j c_j b = \left(\sum_j c_j\right) b = b.$$

Thus every affine combination is another solution. This proves the solution set is affine. If the system is inconsistent, its solution set is empty and the closure assertion holds vacuously; if it has a solution p, the more concrete form is $p + \operatorname{Nul} A$.

Exercise 8.1.16: Affine closure, flats, and hull identities

For points x_j satisfying $x_j \cdot v = k$ and coefficients summing to 1,

$$\left(\sum_j c_j x_j\right) \cdot v = \sum_j c_j (x_j \cdot v) = \sum_j c_j k = k.$$

Hence the set is closed under affine combinations. If $v = 0$, the set is all of $\mathbb{R}^n$ when $k = 0$, and empty otherwise; both satisfy that closure convention.

Exercise 8.1.17: Affine closure, flats, and hull identities

Take $S = \{(0, 0, 5), (1, 0, 5), (0, 1, 5)\}$. Their differences from the first point are e_1, e_2, which span all horizontal directions. Thus

$$\operatorname{aff} S = (0, 0, 5) + \operatorname{Span}\{e_1, e_2\} = \{(x_1, x_2, 5) : x_1, x_2 \in \mathbb{R}\}.$$

This is exactly the required plane, not merely a subset of it.

Exercise 8.1.18: Affine closure, flats, and hull identities

Choose $p = (6, 0, 0)$, $u = (-1, 2, 0)$, $v = (3, 0, 2)$, and the four distinct points

$$S = \{p, p + u, p + v, p + u + v\} = \{(6, 0, 0), (5, 2, 0), (9, 0, 2), (8, 2, 2)\}.$$

Each satisfies $2x_1 + x_2 - 3x_3 = 12$. The two directions u, v are independent, so the affine hull is the plane $p + \operatorname{Span}\{u, v\}$. Conversely, a point on the required plane has the expression

$$p + \frac{x_2}{2}u + \frac{x_3}{2}v = \left(6 - \frac{x_2}{2} + \frac{3x_3}{2}, x_2, x_3\right),$$

and the first coordinate equals x_1 by its plane equation. Thus no points of the desired plane are missing from the hull.

Exercise 8.1.19: Affine closure, flats, and hull identities

Take finitely many points $y_j = f(x_j)$ in $f(S)$ with $x_j \in S$, and coefficients summing to 1. Since S is affine, $x = \sum_j c_j x_j$ is in S. Linearity gives $\sum_j c_j y_j = f(x)$, which is in $f(S)$. Thus $f(S)$ is closed under affine combinations and is affine. If S is empty, its image is empty and the same closure statement is vacuous.

Exercise 8.1.20: Affine closure, flats, and hull identities

For points $x_j \in S$ we know $f(x_j) \in T$. If $\sum_j c_j = 1$, linearity gives $f(\sum_j c_j x_j) = \sum_j c_j f(x_j)$. Because T is affine, this image belongs to T, hence the combined input belongs to S. This verifies affine closure of the preimage, including the empty-preimage case.

Exercise 8.1.21: Affine closure, flats, and hull identities

Every point of aff A is an affine combination of finitely many points of A. They also lie in B because $A \subseteq B$. Since B is affine, the combination lies in B. Therefore aff $A \subseteq B$.

Exercise 8.1.22: Affine closure, flats, and hull identities

The affine set aff B contains B and hence contains A. Applying Exercise 21 with that affine set gives aff $A \subseteq$ aff B.

Exercise 8.1.23: Affine closure, flats, and hull identities

Both $A \subseteq A \cup B$ and $B \subseteq A \cup B$. Exercise 22 therefore gives $\operatorname{aff} A \subseteq \operatorname{aff}(A \cup B)$ and $\operatorname{aff} B \subseteq \operatorname{aff}(A \cup B)$. Taking their union proves the requested inclusion.

Exercise 8.1.24: Affine closure, flats, and hull identities

Take $A = \{(0,0)\}$ and $B = \{(1,0)\}$. Each singleton is its own affine hull, so their hull union is just two points. In contrast, $\operatorname{aff}(A \cup B) = \{(t,0) : t \in \mathbb{R}\}$ is an entire line. For instance $(1/2, 0)$ lies in the latter but not the former, proving strict inclusion.

Exercise 8.1.25: Affine closure, flats, and hull identities

Since $A \cap B \subseteq A$ and $A \cap B \subseteq B$, monotonicity of affine hull gives $\operatorname{aff}(A \cap B) \subseteq \operatorname{aff} A$ and $\operatorname{aff}(A \cap B) \subseteq \operatorname{aff} B$. Membership in both is exactly membership in their intersection, yielding the asserted inclusion.

Exercise 8.1.26: Affine closure, flats, and hull identities

Let $A = \{(0,0), (1,0), (0,1)\}$ and $B = \{(0,0), (2,0), (0,2)\}$. Their intersection is the singleton $\{(0,0)\}$, whose affine hull is itself. Each original set contains three noncollinear points and hence has affine hull $\mathbb{R}^2$. Thus $\operatorname{aff} A \cap \operatorname{aff} B = \mathbb{R}^2$, strictly larger than $\operatorname{aff}(A \cap B) = \{(0,0)\}$. This example does not rely on a convention for the affine hull of the empty set.

8.2 Affine independence

An indexed set of points is affinely dependent if there are coefficients, not all zero, whose sum is 0 and whose weighted point sum is 0. It is affinely independent otherwise. Appending a final coordinate 1 converts this definition into the ordinary null-space test for linear dependence.

An affinely independent set gives *unique* affine coefficients, called barycentric coordinates, for every point in its affine hull. These coefficients sum to 1, but may be negative outside the convex region between the points. For a triangle, the coordinate belonging to a vertex is zero on the line through the other two vertices, positive on the vertex's side of that line, and negative on the opposite side. This geometric sign rule is particularly useful in Exercises 15 and 16.

Practice problem 8.2.P1: A fast collinearity test

For points v_1, v_2, v_3, compute the two differences $u = v_2 - v_1$ and $w = v_3 - v_1$. They are collinear exactly when u, w are linearly dependent, which can be checked by a two-column row reduction. In $\mathbb{R}^2$ one can simply test $\det[u\ w] = 0$. Repeated points also satisfy the test. If the differences are independent, the three points are not on one line and are affinely independent.

Practice problem 8.2.P2: A nontrivial relation with coefficient sum zero

Append a final 1 to each point and form

$$H = \begin{bmatrix} 4 & 1 & 5 & 1 \\ 1 & 0 & 4 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

An affine dependence is exactly a nonzero solution of $Hc = 0$: its first rows assert $\sum c_j v_j = 0$, and its last row asserts $\sum c_j = 0$. Row reduction gives

$$\operatorname{rref} H = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & \frac{5}{4} \\ 0 & 0 & 1 & \frac{3}{4} \end{bmatrix}.$$

There is a free variable, so the points are affinely dependent. One explicit nonzero coefficient column is

$$c = \begin{bmatrix} 4 \\ -5 \\ -3 \\ 4 \end{bmatrix}.$$

Its entries sum to zero and multiplying Hc gives zero in every coordinate. Thus

$$(4)v_1 + (-5)v_2 + (-3)v_3 + (4)v_4 = 0$$

is the requested affine dependence relation. The weights sum to 0, not 1, for a dependence.

Exercise 8.2.1: Test affine dependence using homogeneous columns

Append a final 1 to each point and form

$$H = \begin{bmatrix} 3 & 0 & 2 \\ -3 & 6 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

An affine dependence is exactly a nonzero solution of $Hc = 0$: its first rows assert $\sum c_j v_j = 0$, and its last row asserts $\sum c_j = 0$. Row reduction gives

$$\operatorname{rref} H = \begin{bmatrix} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{3} \\ 0 & 0 & 0 \end{bmatrix}.$$

There is a free variable, so the points are affinely dependent. One explicit nonzero coefficient column is

$$c = \begin{bmatrix} -2 \\ -1 \\ 3 \end{bmatrix}.$$

Its entries sum to zero and multiplying Hc gives zero in every coordinate. Thus

$$(-2)v_1 + (-1)v_2 + (3)v_3 = 0$$

is the requested affine dependence relation. The weights sum to 0, not 1, for a dependence.

Exercise 8.2.2: Test affine dependence using homogeneous columns

Append a final 1 to each point and form

$$H = \begin{bmatrix} 2 & 5 & -3 \\ 1 & 4 & -2 \\ 1 & 1 & 1 \end{bmatrix}.$$

An affine dependence is exactly a nonzero solution of $Hc = 0$: its first rows assert $\sum c_j v_j = 0$, and its last row asserts $\sum c_j = 0$. Row reduction gives

$$\operatorname{rref} H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Every column is a pivot column, so $Hc = 0$ has only the zero solution. Hence the points are affinely independent and no affine dependence relation exists.

Exercise 8.2.3: Test affine dependence using homogeneous columns

Append a final 1 to each point and form

$$H = \begin{bmatrix} 1 & -2 & 2 & 0 \\ 2 & -4 & -1 & 15 \\ -1 & 8 & 11 & -9 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

An affine dependence is exactly a nonzero solution of $Hc = 0$: its first rows assert $\sum c_j v_j = 0$, and its last row asserts $\sum c_j = 0$. Row reduction gives

$$\operatorname{rref} H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Every column is a pivot column, so $Hc = 0$ has only the zero solution. Hence the points are affinely independent and no affine dependence relation exists.

Exercise 8.2.4: Test affine dependence using homogeneous columns

Append a final 1 to each point and form

$$H = \begin{bmatrix} -2 & 0 & 1 & -2 \\ 5 & -3 & -2 & 7 \\ 3 & 7 & -6 & -3 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

An affine dependence is exactly a nonzero solution of $Hc = 0$: its first rows assert $\sum c_j v_j = 0$, and its last row asserts $\sum c_j = 0$. Row reduction gives

$$\operatorname{rref} H = \begin{bmatrix} 1 & 0 & 0 & \frac{6}{5} \\ 0 & 1 & 0 & -\frac{3}{5} \\ 0 & 0 & 1 & \frac{2}{5} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There is a free variable, so the points are affinely dependent. One explicit nonzero coefficient column is

$$c = \begin{bmatrix} -6 \\ 3 \\ -2 \\ 5 \end{bmatrix}.$$

Its entries sum to zero and multiplying Hc gives zero in every coordinate. Thus

$$(-6)v_1 + (3)v_2 + (-2)v_3 + (5)v_4 = 0$$

is the requested affine dependence relation. The weights sum to 0, not 1, for a dependence.

Exercise 8.2.5: Test affine dependence using homogeneous columns

Append a final 1 to each point and form

$$H = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 5 & 5 \\ -2 & 1 & 1 & -3 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

An affine dependence is exactly a nonzero solution of $Hc = 0$: its first rows assert $\sum c_j v_j = 0$, and its last row asserts $\sum c_j = 0$. Row reduction gives

$$\operatorname{rref} H = \begin{bmatrix} 1 & 0 & 0 & \frac{4}{3} \\ 0 & 1 & 0 & -\frac{5}{3} \\ 0 & 0 & 1 & \frac{4}{3} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There is a free variable, so the points are affinely dependent. One explicit nonzero coefficient column is

$$c = \begin{bmatrix} -4 \\ 5 \\ -4 \\ 3 \end{bmatrix}.$$

Its entries sum to zero and multiplying Hc gives zero in every coordinate. Thus

$$(-4)v_1 + (5)v_2 + (-4)v_3 + (3)v_4 = 0$$

is the requested affine dependence relation. The weights sum to 0, not 1, for a dependence.

Exercise 8.2.6: Test affine dependence using homogeneous columns

Append a final 1 to each point and form

$$H = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 3 & -1 & 5 & 5 \\ 1 & -2 & 2 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

An affine dependence is exactly a nonzero solution of $Hc = 0$: its first rows assert $\sum c_j v_j = 0$, and its last row asserts $\sum c_j = 0$. Row reduction gives

$$\operatorname{rref} H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Every column is a pivot column, so $Hc = 0$ has only the zero solution. Hence the points are affinely independent and no affine dependence relation exists.

Exercise 8.2.7: Unique barycentric coordinates in a higher-dimensional affine hull

To require both the point equation and the coefficient-sum condition, solve

$$Hc = \widetilde{y}, \qquad H = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 1 & 2 \\ 2 & 0 & -2 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad \widetilde{y} = \begin{bmatrix} 5 \\ 4 \\ -2 \\ 2 \\ 1 \end{bmatrix}.$$

The last row of H enforces $c_1 + \cdots + c_k = 1$. Row reduction of the augmented system gives

$$\operatorname{rref}[H \mid \widetilde{y}] = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

One affine coefficient column is

$$c = \begin{bmatrix} -2 \\ 4 \\ -1 \end{bmatrix}.$$

Its entries sum to 1, and substituting them in $\sum c_j v_j$ gives exactly $\begin{bmatrix} 5 \\ 4 \\ -2 \\ 2 \end{bmatrix}$. Thus the requested affine combination exists. Since the points are affinely independent, the coefficient column is unique and is precisely the requested barycentric coordinate column.

Exercise 8.2.8: Unique barycentric coordinates in a higher-dimensional affine hull

To require both the point equation and the coefficient-sum condition, solve

$$Hc = \widetilde{y}, \qquad H = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 4 \\ -2 & 0 & -6 \\ 1 & 2 & 5 \\ 1 & 1 & 1 \end{bmatrix}, \quad \widetilde{y} = \begin{bmatrix} -1 \\ 1 \\ -4 \\ 0 \\ 1 \end{bmatrix}.$$

The last row of H enforces $c_1 + \cdots + c_k = 1$. Row reduction of the augmented system gives

$$\operatorname{rref}[H \mid \widetilde{y}] = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

One affine coefficient column is

$$c = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}.$$

Its entries sum to 1, and substituting them in $\sum c_j v_j$ gives exactly $\begin{bmatrix} -1 \\ 1 \\ -4 \\ 0 \end{bmatrix}$. Thus the requested affine combination exists. Since the points are affinely independent, the coefficient column is unique and is precisely the requested barycentric coordinate column.

Exercise 8.2.9: Affine independence and barycentric meaning

(a) **True.** The differences are based at v_2, not v_1, but the same criterion works for any base point. If $\sum_{j\neq 2} c_j(v_j - v_2) = 0$ is nontrivial, give v_2 coefficient $-\sum_{j\neq 2} c_j$. The resulting nontrivial point relation has coefficient sum zero.

(b) **False.** Independent homogeneous columns mean the original points are affinely *independent*, not dependent.

(c) **False.** A dependence requires coefficient sum 0. The distinct points $v_1 = (1, 0)$, $v_2 = (2, 0)$ are affinely independent, but $2v_1 - v_2 = 0$ has coefficient sum 1, satisfying the proposed incorrect test.

(d) **False.** A negative coordinate can occur for a point in the affine hull, but outside the convex hull. For the two points $(0, 0), (1, 0)$, the point $(2, 0)$ has barycentric coordinates $(-1, 2)$ and is still in their affine line.

(e) **True for the stated triangle.** A point in a closed triangular region is a convex combination of the vertices, so its barycentric coordinates are nonnegative; an edge or vertex can give zeros. The ray is merely a way to locate that point and does not change this condition.

Exercise 8.2.10: Affine independence and barycentric meaning

(a) **False.** A nontrivial affine dependence gives a nontrivial linear dependence of the homogeneous forms with the very same coefficients. They cannot be independent.

(b) **True.** Three independent differences based at v_1 imply affine independence of the four original points by the difference criterion.

(c) **False without affine independence.** For $S = \{(1, 0), (2, 0), (3, 0)\}$, the middle point is both itself and $\frac{1}{2}(1, 0) + \frac{1}{2}(3, 0)$. These are distinct affine representations.

(d) **True as an affine color rule.** Use the three barycentric weights for each color component at the vertices. Inside the triangle this interpolates the colors; outside, negative weights give affine extrapolation and may leave a physical color range.

(e) **True.** A point on an edge has coefficient zero for the opposite vertex. Hence the three coefficients are not all strictly positive, including the vertex endpoints of the edge.

Exercise 8.2.11: Affine independence and barycentric meaning

Append a 1 to every point of $\mathbb{R}^3$, producing vectors in $\mathbb{R}^4$. Five or more vectors in a four-dimensional vector space are linearly dependent. Therefore some nontrivial coefficient column has both weighted point sum zero and coefficient sum zero. That is an affine dependence of the original indexed points.

Exercise 8.2.12: Affine independence and barycentric meaning

The homogeneous forms lie in $\mathbb{R}^{n+1}$. If $p \geq n + 2$, there are more columns than the dimension, so their homogeneous system has a free variable and a nonzero solution c. Its first n equations give $\sum_j c_j v_j = 0$, and its last gives $\sum_j c_j = 0$. This is precisely the affine-dependence definition.

Exercise 8.2.13: Affine independence and barycentric meaning

Suppose $c_1v_1 + c_2v_2 = 0$, $c_1 + c_2 = 0$, and the coefficient pair is nonzero. Then $c_2 = -c_1$, with $c_1 \neq 0$. The point equation becomes $c_1(v_1 - v_2) = 0$, so division by c_1 gives $v_1 = v_2$. Conversely, if $v_1 = v_2$, take $(c_1, c_2) = (1, -1)$. Their sum is zero and their weighted point sum is zero, so the indexed pair is affinely dependent. Both directions use only the definition.

Exercise 8.2.14: Affine independence and barycentric meaning

For $S_1 = \{(1,0), (2,0), (3,0)\}$, the relation $v_1 - 2v_2 + v_3 = 0$ has nonzero coefficients summing to zero, so this set is both affinely and linearly dependent.

For $S_2 = \{(1,0), (2,0)\}$, the relation $2v_1 - v_2 = 0$ proves linear dependence, but the two points are distinct and hence affinely independent by Exercise 13. All points in both examples are nonzero, as requested. Linear dependence does not force its coefficients to sum to zero.

Exercise 8.2.15: Barycentric coordinates and signs in the printed figure

(a) Check affine independence.

The homogeneous matrix has determinant $-8 \neq 0$, so its three columns are independent and the original points are affinely independent.

$$H = \begin{bmatrix} -1 & 0 & 2 \\ 2 & 4 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

(b) Solve for all five ordered coordinate triples.

The inverse mapping, obtained by row reduction, is

$$H^{-1} = \begin{bmatrix} -\frac{1}{2} & -\frac{1}{4} & 1 \\ \frac{1}{4} & \frac{3}{8} & -\frac{1}{2} \\ \frac{1}{4} & -\frac{1}{8} & \frac{1}{2} \end{bmatrix}.$$

Thus multiply $H^{-1}(x, y, 1)^T$ for each given point. The exact results are

point	r	s	t
p_1	$-\frac{3}{4}$	$\frac{9}{8}$	$\frac{5}{8}$
p_2	0	$\frac{1}{2}$	$\frac{1}{2}$
p_3	$\frac{7}{4}$	$-\frac{5}{8}$	$-\frac{1}{8}$
p_4	$\frac{3}{4}$	$-\frac{5}{8}$	$\frac{7}{8}$
p_5	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{5}{8}$

Each row sums to 1 and reconstructs its point, supplying independent checks on the arithmetic.

(c) Use the positions in Figure 8 rather than inventing coordinates for its unlabeled points.

A coordinate is zero on the opposite edge line, positive on its own vertex's side, and negative on the other side. Reading the three side tests in the figure gives

$$p_6 : \ (-,-,+), \qquad p_7 : \ (0,+,-), \qquad p_8 : \ (+,+,-).$$

The zero for p_7 matters: the dot is on the *extended line* through v_2, v_3, beyond v_2, not in an adjacent open region. Its second weight is positive and third negative along that extension. Point p_6 lies across both edge lines opposite v_1 and v_2; point p_8 is opposite v_3 but on the other two vertices' positive sides.

Exercise 8.2.16: A plotted triangle with extended sides and boundary points

(a) Verify independence of the triangle vertices.

The homogeneous matrix

$$H = \begin{bmatrix} 0 & 1 & 4 \\ 1 & 5 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$

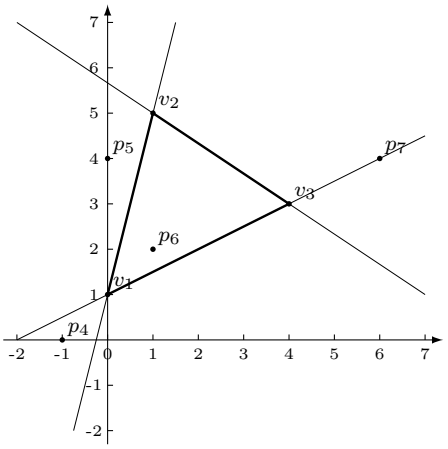
has determinant $-14 \neq 0$, proving affine independence.

(b) Compute the first three coordinate triples.

Solve $Hc = (p_x, p_y, 1)^T$. Row reduction gives

point	r	s	t
p_1	$-\frac{2}{7}$	$\frac{5}{7}$	$\frac{4}{7}$
p_2	$\frac{2}{7}$	$-\frac{5}{7}$	$\frac{10}{7}$
p_3	$\frac{2}{7}$	$\frac{2}{7}$	$\frac{3}{7}$

(c) Plot the remaining points and test which side of each line they occupy.



The thin lines extend the three sides; the bold triangle joins the actual vertices. For a weight belonging to v_j, compare the point with v_j across the line through the other two vertices. These side tests give

$$p_4 : \ (+,-,-), \qquad p_5 : \ (+,+,-), \qquad p_6 : \ (+,+,+), \qquad p_7 : \ (-,0,+).$$

Here p_6 lies inside the triangle, while $p_7 = (6, 4)$ lies exactly on the extension of the side through $v_1 = (0, 1)$ and $v_3 = (4, 3)$, so its v_2 coordinate is zero. This boundary check prevents treating every exterior point as having three nonzero coordinates. No numerical barycentric weights are needed for these sign conclusions.

Exercise 8.2.17: Barycentric proofs and geometric consequences

What we must show. Prove both existence and uniqueness in Theorem 6. Existence is the definition of $p \in \operatorname{aff} S$: some coefficients c_j sum to 1 and satisfy $p = \sum_j c_j v_j$. For uniqueness suppose also $p = \sum_j d_j v_j$ with $\sum_j d_j = 1$. Subtract the two point equations and the two coefficient sums:

$$\sum_j (c_j - d_j) v_j = 0, \qquad \sum_j (c_j - d_j) = 0.$$

Affine independence says the only coefficient column satisfying both zero equations is zero. Thus $c_j - d_j = 0$ for every j, so all weights agree. This is uniqueness of the entire ordered coordinate column, not merely uniqueness of the represented point.

Exercise 8.2.18: Barycentric proofs and geometric consequences

The ordered vertices are $ae_1, be_2, ce_3, 0$, with $a, b, c > 0$ in the stated standard position. If $p = (p_1, p_2, p_3)$ has barycentric coordinates (r, s, t, u), coordinate comparison gives $ar = p_1$, $bs = p_2$, and $ct = p_3$. The sum condition then determines the last coordinate. Hence

$$(r, s, t, u) = \left(\frac{p_1}{a}, \frac{p_2}{b}, \frac{p_3}{c}, 1 - \frac{p_1}{a} - \frac{p_2}{b} - \frac{p_3}{c} \right).$$

These formulas work for every point of $\mathbb{R}^3$, not only interior points. The point belongs to the closed tetrahedron exactly when all four coordinates are nonnegative. The three edge vectors ae_1, be_2, ce_3 are independent, so the coordinate column is unique.

Exercise 8.2.19: Barycentric proofs and geometric consequences

Choose a nontrivial affine-dependence coefficient triple c_1, c_2, c_3. By assumption $\sum_j c_j p_j = 0$ and $\sum_j c_j = 0$. Linearity gives

$$\sum_j c_j f(p_j) = f\left(\sum_j c_j p_j \right) = f(0) = 0.$$

The coefficients have not changed and are still not all zero, so the indexed image points satisfy an affine dependence. The map may identify some points; this does not destroy the displayed indexed relation.

Exercise 8.2.20: Barycentric proofs and geometric consequences

Suppose the translated points have a coefficient relation with $\sum_j c_j = 0$ and $\sum_j c_j (p_j + q) = 0$. Expanding yields

$$0 = \sum_j c_j p_j + \left(\sum_j c_j \right) q = \sum_j c_j p_j.$$

Affine independence of the original points now forces every $c_j = 0$. Therefore no nontrivial affine dependence exists among the translated points. Translation changes location but not affine independence.

Exercise 8.2.21: Barycentric proofs and geometric consequences

The homogeneous-column determinant is

$$D = \det \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ 1 & 1 & 1 \end{bmatrix}.$$

Subtract its first column from columns 2 and 3, then expand across the final row. This gives $D = \det[b-a\ c-a]$. The parallelogram on those two edges has area $|D|$, and the triangle has half that area. The stipulated vertex order makes $D > 0$, so the triangle area is $D/2$ without absolute-value bars. For a different order one would need $|D|/2$.

Exercise 8.2.22: Barycentric proofs and geometric consequences

A point on the line is $p = (1-t)a + tb$ for some real t. Because these two weights sum to 1, its homogeneous form satisfies $\widetilde{p} = (1-t)\widetilde{a} + t\widetilde{b}$. Therefore the three homogeneous columns are linearly dependent, and $\det[\widetilde{a}\ \widetilde{b}\ \widetilde{p}] = 0$. This also follows by expanding linearly in the third column, where both resulting determinants have repeated columns.

Exercise 8.2.23: Barycentric proofs and geometric consequences

Write $\widetilde{p} = r\widetilde{a} + s\widetilde{b} + t\widetilde{c}$ and let $D = \det[\widetilde{a}\ \widetilde{b}\ \widetilde{c}] > 0$. Expand a determinant linearly in its replaced column. Repeated-column terms vanish, so

$$\det[\widetilde{p}\ \widetilde{b}\ \widetilde{c}] = rD, \quad \det[\widetilde{a}\ \widetilde{p}\ \widetilde{c}] = sD, \quad \det[\widetilde{a}\ \widetilde{b}\ \widetilde{p}] = tD.$$

Since p is interior, it lies on the same side of each opposite edge as the replaced vertex; each displayed subtriangle has the same positive orientation as the original triangle. Thus its ordinary area is one half of its displayed determinant. Divide those area expressions by $D/2$, the area of the full triangle, to obtain

$$r = \frac{\text{area}(\triangle pbc)}{\text{area}(\triangle abc)}, \qquad s = \frac{\text{area}(\triangle apc)}{\text{area}(\triangle abc)},$$

$$t = \frac{\text{area}(\triangle abp)}{\text{area}(\triangle abc)}.$$

These are positive ratios for an interior point. Signed determinants, rather than unsigned areas alone, are needed if extending the formulas outside the triangle.

Exercise 8.2.24: Barycentric proofs and geometric consequences

Because q lies on the segment bc, it has the form $q = (1-t)b + tc$ for some $0 \le t \le 1$. The given line point therefore has affine representation

$$p = xa + (1-x)(1-t)b + (1-x)tc.$$

Its three weights sum to 1. Appending final coordinates and expanding the determinant linearly in its first column gives

$$\det[\widetilde{p}\ \widetilde{b}\ \widetilde{c}] = x \det[\widetilde{a}\ \widetilde{b}\ \widetilde{c}],$$

because both terms involving $\widetilde{b}$ or $\widetilde{c}$ in the first column have repeated columns. The original triangle is nondegenerate, so its vertices are affinely independent. Uniqueness of barycentric coordinates makes x exactly the first coordinate of p.

Also $p - q = x(a - q)$, so for $0 \le x \le 1$, $\|p - q\|/\|a - q\| = x$. For other real x, it is an *oriented* relative position; the ordinary distance ratio is $|x|$. A vertex color and the color at q combine as $xC_a + (1 - x)C_q$, varying linearly and continuously as the point moves. This is the stated smooth interpolation along the segment.

8.3 Convex combinations

A linear combination allows arbitrary coefficients. An affine combination requires that their sum be 1. A convex combination requires both sum 1 and *nonnegative* coefficients. Thus $\operatorname{conv} S \subseteq \operatorname{aff} S \subseteq \operatorname{Span} S$. For an affinely independent list, its affine coordinates are unique: a negative coordinate rules out convex membership, all positive coordinates put the point in the relative interior of its simplex, and zero coordinates locate it on a face. A convex hull consists of finite convex combinations; it need not include limit points not obtained by those combinations.

Practice problem 8.3.P1: Test a triangle in three-dimensional space

Use the supplied vertices $v_1 = (6, 2, 2)$, $v_2 = (7, 1, 5)$, $v_3 = (-2, 4, -1)$. To find affine coordinates solve $p - v_1 = a(v_2 - v_1) + b(v_3 - v_1)$, then use coefficients $(1 - a - b, a, b)$. For $p_1 = (1, 3, 1)$ the solution is $a = 1/3, b = 2/3$; checking gives $p_1 = 0v_1 + (1/3)v_2 + (2/3)v_3$. All coefficients are nonnegative, so $p_1 \in \operatorname{conv} S$. For $p_2 = (3, 2, 1)$ the first two coordinate equations give $a = 1, b = 1/2$, but the third would require $3a - 3b = -1$, whereas it equals $3/2$. Thus p_2 is not even in the affine hull and cannot be in the convex hull.

Practice problem 8.3.P2: The hull of the positive reciprocal curve

What we must show. Describe the hull of $S = \{(x, 1/x) : x > 0\}$ exactly. No convex combination has $x \le 0$. Moreover convexity of $1/x$ on positive inputs gives $\sum_i \lambda_i/x_i \ge 1/(\sum_i \lambda_i x_i)$. For example this follows directly from Cauchy–Schwarz applied to $\sqrt{\lambda_i x_i}$ and $\sqrt{\lambda_i/x_i}$. Hence the hull lies in $\{(x, y) : x > 0, y \ge 1/x\}$. Conversely, for a point (x, y) in that set draw the line of slope -1 through it, $u + v = x + y$. Its intersections with $uv = 1$ have positive x-coordinates given by roots of $u^2 - (x + y)u + 1 = 0$. The polynomial at $u = x$ is $1 - xy \le 0$, so x lies between the two roots. Thus (x, y) lies on the line segment joining these two curve points, or is itself the single tangency point. It is a convex combination of them. Therefore the hull is exactly the region on and above the curve.

Exercise 8.3.1: A triangle with a missing upper edge

Every convex combination of vertical-segment points and $(2, 0)$ can be written $(1 - \lambda)(0, a) + \lambda(2, 0)$ with $0 \le \lambda \le 1$ and $0 \le a < 1$. (Combine all the vertical points into their weighted average first.) Thus the hull is

$$\{(x, y) : 0 \le x < 2,\ 0 \le y < 1 - x/2\} \ \cup\ \{(2, 0)\}.$$

Conversely any point in the first set uses $\lambda = x/2$ and $a = y/(1 - x/2)$, which satisfy the required inequalities. The lower edge and right vertex are included, but the sloping upper edge and the point $(0, 1)$ are excluded. Replacing the strict inequality by a weak one would add points not actually in the convex hull.

Exercise 8.3.2: Three convex hulls of graphs

(a) The reciprocal graph starting at $x = 1/2$.

The exact hull is

$$\{(1/2, 2)\} \ \cup\ \{(x, y) : x > 1/2,\ 1/x \le y < 2\}.$$

Every combination has $x \geq 1/2$, lies on or above the reciprocal curve by convexity, and has $y \leq 2$. Equality $y = 2$ forces every positive-weight input to be the endpoint $(1/2, 2)$, so it occurs only there. For the converse fix $x > 1/2$ and $1/x \leq y < 2$. Put $b = (2x-1)/(2-y)$. The lower bound on y gives $b \geq x$. The chord joining $(1/2, 2)$ and $(b, 1/b)$ has height $2 - (2x-1)/b = y$ at horizontal coordinate x. Because $1/2 < x \leq b$, the point lies on that segment and is in the hull.

(b) The sine graph on the whole real line.

The hull is the closed strip $\{(x, y) : -1 \leq y \leq 1\}$. The range of sine bounds every combination within the strip. Conversely its maximum points $(\pi/2 + 2j\pi, 1)$ extend in both horizontal directions. For any fixed real x, two such points bracket x, so their segment contains $(x, 1)$. The minimum points similarly give $(x, -1)$. The segment between these two hull points contains every (x, y) in the strip.

(c) The square-root graph.

The hull is

$$\{(0,0)\} \ \cup \ \{(x,y) : x > 0, \ 0 < y \leq \sqrt{x}\}.$$

Concavity of square root bounds the height of a convex combination by $\sqrt{x}$; nonnegativity is automatic. If $x > 0$, some positive-weight input has positive x and therefore positive height, so height cannot be zero. Conversely, for $x > 0$ and $0 < y \leq \sqrt{x}$, take $a = (x/y)^2$ and $\lambda = y^2/x \in (0, 1]$. Then $\lambda(a, \sqrt{a}) + (1-\lambda)(0,0) = (x, y)$. This proves the formula and explains why the positive horizontal axis is not included.

Exercise 8.3.3: Use the unique coordinates from Section 8.1, Exercise 5

The three orthogonal generators are $b_1 = (2, 1, 1)$, $b_2 = (1, 0, -2)$, $b_3 = (2, -5, 1)$. Their squared lengths are $6, 5, 30$, so the unique linear coefficients of p are $c_j = (p \cdot b_j)/(b_j \cdot b_j)$. For the three supplied points the results are

p	(c_1, c_2, c_3)	sum
$(3, 8, 4)$	$(3, -1, -1)$	1
$(6, -3, 3)$	$(2, 0, 1)$	3
$(0, -1, -5)$	$(-1, 2, 0)$	1

The first and third have negative coordinates; the second does not even have coefficient sum one. Since the linear representation in a basis is unique, no alternative nonnegative sum-one representation exists.

Conclusion. None of the three points is in $\operatorname{conv} S$.

Exercise 8.3.4: A second list with the same orthogonal generators

The same dot-product calculation gives

p	(c_1, c_2, c_3)	sum
$(0, -19, -5)$	$(-4, 2, 3)$	1
$(3/2, -13/10, -1/2)$	$(1/5, 1/2, 3/10)$	1
$(5, -4, 0)$	$(1, 1, 1)$	3

Only the middle row has all coefficients nonnegative and sum one. Thus $p_2 \in \operatorname{conv} S$, while p_1, p_3 are not.

Exercise 8.3.5: Convex coordinates in a tetrahedron

Append a final coordinate 1 to each of the four supplied v-vectors. Solving the resulting 4×4 systems for the two points gives

$$[p_1]_S = (-1/6, 1/3, 2/3, 1/6), \qquad [p_2]_S = (1/3, 1/3, 1/6, 1/6).$$

Here coordinates refer to affine/barycentric coordinates. The appended matrix is invertible, so these representations are unique. For instance $(1/3)v_1 + (1/3)v_2 + (1/6)v_3 + (1/6)v_4 = (0, -2, 2) = p_2$. The first representation has a negative coefficient and therefore is not convex; the second has all coefficients positive and sum one.

Conclusion. p_1 is outside the convex hull; p_2 is in it.

Exercise 8.3.6: Distinguish the linear, affine, and convex hulls

The supplied orthogonal vectors are $v_1 = (2, 0, -1, 2)$, $v_2 = (0, -2, 2, 1)$, $v_3 = (-2, 1, 0, 2)$; each has squared length 9. For a proposed p, first compute $c_j = (p \cdot v_j)/9$ and check whether $\sum c_j v_j = p$. Only after this check do sum-one and sign tests apply.

(a) For $p_1 = (-1, 2, -3/2, 5/2)$, the coefficients are $(1/2, -1/2, 1)$ and reconstruct p_1. They sum to one but include a negative coefficient. Thus p_1 is in the span and affine hull, but not the convex hull.

(b) For $p_2 = (-1/2, 0, 1/4, 7/4)$, the coefficients are $(1/4, 1/4, 1/2)$. They reconstruct p_2, sum to one, and are nonnegative. Thus p_2 is in all three hulls.

(c) For $p_3 = (6, -4, 1, -1)$, the coefficients are $(1, 1, -2)$ and reconstruct p_3, but their sum is zero. Hence p_3 is in the span only, not the affine or convex hull.

(d) For $p_4 = (-1, -2, 0, 4)$, the dot-product coefficients are $(2/3, 8/9, 8/9)$, but the reconstruction error is $p_4 - \sum c_j v_j = (-5/9, -10/9, -10/9, 0) \neq 0$. Thus p_4 is not in the span and consequently in neither smaller hull.

Exercise 8.3.7: Barycentric coordinates locate four planar points

(a) Solve the affine-coordinate systems.

With $T = \{(-1, 0), (2, 3), (4, 1)\}$, solve

$$\begin{pmatrix} -1 & 2 & 4 \\ 0 & 3 & 1 \\ 1 & 1 & 1 \end{pmatrix} c = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}.$$

The four answers, in the listed order, are

$$(1/3, 1/6, 1/2), \quad (0, 1/2, 1/2), \quad (1/2, -1/4, 3/4), \quad (1/2, 3/4, -1/4).$$

(b) Use signs and zero entries.

p_1 has all positive entries and is inside the triangle. p_2 has one zero and the other entries positive, so it is on the edge joining vertices two and three. p_3, p_4 have negative entries and are outside.

Exercise 8.3.8: Another barycentric-coordinate table

For $T = \{(2,0),(0,5),(-1,1)\}$, use the appended-coordinate system

$$\begin{pmatrix} 2 & 0 & -1 \\ 0 & 5 & 1 \\ 1 & 1 & 1 \end{pmatrix} c = (x, y, 1)^T.$$

For p_1, p_2, p_3, p_4 its solutions are respectively

$$(12/13, 3/13, -2/13), \quad (8/13, 2/13, 3/13), \quad (2/3, 0, 1/3), \quad (9/13, -1/13, 5/13).$$

Thus p_1 is outside, p_2 inside, p_3 on the edge joining vertices one and three, and p_4 outside. The p_3 supplied in the exercise is $(1, 1/3)$, not $(1, 1)$; preserving this fraction is essential.

Exercise 8.3.9: Read positions from tetrahedral coordinates

A point is inside precisely when all four barycentric coordinates are positive. Nonnegative coordinates with a zero put it on a surface; two zeros put it on an edge. A negative coordinate puts it outside. Therefore $p_1 = (2, 0, 0, -1)$ is outside; $p_2 = (0, 1/2, 1/4, 1/4)$ is on the face opposite vertex one, but not an edge; $p_3 = (1/2, 0, 3/2, -1)$ is outside; $p_4 = (1/3, 1/4, 1/4, 1/6)$ is inside; and $p_5 = (1/3, 0, 2/3, 0)$ is on the edge joining vertices one and three.

Exercise 8.3.10: A second tetrahedral list

Using the same criteria, $q_1 = (1/8, 1/4, 1/8, 1/2)$ is inside; $q_2 = (3/4, -1/4, 0, 1/2)$ is outside; $q_3 = (0, 3/4, 1/4, 0)$ is on the edge joining vertices two and three; $q_4 = (0, -2, 0, 3)$ is outside; and $q_5 = (1/3, 1/3, 1/3, 0)$ is on the face opposite vertex four, not an edge.

Exercise 8.3.11: Three statements about convexity

(a) **False.** Sum one makes a combination affine, but negative coefficients are still possible. Convexity also requires each coefficient nonnegative.

(b) **False.** A singleton $S = \{v\}$ has $\operatorname{conv} S = S$. More generally this equality holds for any convex S.

(c) **False.** Two singleton sets are convex but their union $\{(0,0),(2,0)\}$ misses the midpoint $(1,0)$.

Exercise 8.3.12: Three more statements about hulls

(a) **True.** Containing every segment between two of its points is exactly the definition of convexity.

(b) **True.** If two points belong to both convex sets, their whole joining segment belongs to both, hence to the intersection. The empty intersection is also convex under the usual vacuous definition.

(c) **False.** Carathéodory's theorem says *at most* six points in $\mathbb{R}^5$, not necessarily six distinct available points. A singleton S is an immediate counterexample to the stated requirement.

Exercise 8.3.13: A linear image preserves convexity

For $u = f(x), v = f(y)$ in $f(S)$ and $0 \le t \le 1$, linearity gives $(1-t)u + tv = f((1-t)x + ty)$. Convexity of S puts the input $(1-t)x + ty$ in S, so this output is in $f(S)$. Hence $f(S)$ is convex. Empty sets cause no exception.

Exercise 8.3.14: A linear preimage preserves convexity

For $x, y \in S$, the images $f(x), f(y)$ lie in T. Their combination $(1-t)f(x) + tf(y)$ lies in T for $0 \le t \le 1$ by convexity. This equals $f((1-t)x + ty)$ by linearity, so the input $(1-t)x + ty$ lies in S. Thus the whole segment is in S.

Exercise 8.3.15: Eliminate a vertex in two different ways

Direct coordinate addition checks both supplied identities: $p = (1/3)v_1 + (1/3)v_2 + (1/6)v_3 + (1/6)v_4 = (2, 1)$ and $v_1 - v_2 + v_3 - v_4 = 0$. Add s times the zero relation to the coefficients. They become

$$(1/3 + s,\ 1/3 - s,\ 1/6 + s,\ 1/6 - s).$$

Their sum stays one and their combination stays p. For $s = 1/6$ and $s = -1/6$ all entries are nonnegative and one is zero, giving respectively

$$p = \tfrac{1}{2}v_1 + \tfrac{1}{6}v_2 + \tfrac{1}{3}v_3, \qquad p = \tfrac{1}{6}v_1 + \tfrac{1}{2}v_2 + \tfrac{1}{3}v_4.$$

This is the coefficient-elimination step in the proof of Carathéodory's theorem.

Exercise 8.3.16: A second coefficient-elimination example

The printed instruction says "Repeat Exercise 9," but the supplied convex combination and zero relation clearly call for the procedure of Exercise 15. We carry out that intended task explicitly. The given coefficients are $(1, 72, 37, 11)/121$, whose sum is one; coordinate substitution gives $p = (1, 2)$. The relation $10v_1 - 6v_2 + 7v_3 - 11v_4 = 0$ also has coefficient sum zero. Add $s(10, -6, 7, -11)$ to the original coefficients. For $s = 1/121$, the last coefficient becomes zero and we obtain

$$p = \tfrac{1}{11}v_1 + \tfrac{6}{11}v_2 + \tfrac{4}{11}v_3.$$

For $s = -1/1210$, the first becomes zero and the result is

$$p = \tfrac{3}{5}v_2 + \tfrac{3}{10}v_3 + \tfrac{1}{10}v_4.$$

Each list is nonnegative, sums to one, and reconstructs p.

Exercise 8.3.17: A containing convex set contains the hull

If $A \subseteq B$ and B is convex, then any finite convex combination of points in A is a convex combination of points in B and therefore belongs to B. Every point of $\operatorname{conv} A$ is such a combination, so $\operatorname{conv} A \subseteq B$.

Exercise 8.3.18: Convex hull is monotone under inclusion

If $A \subseteq B$, every convex combination of points in A is also one of points in B. Therefore $\operatorname{conv} A \subseteq \operatorname{conv} B$. This also follows from the preceding exercise because $\operatorname{conv} B$ is convex and contains A.

Exercise 8.3.19: Union of hulls need not fill the combined hull

(a) Prove the inclusion.

Since $A \subseteq A \cup B$ and $B \subseteq A \cup B$, monotonicity gives both $\operatorname{conv} A \subseteq \operatorname{conv}(A \cup B)$ and the corresponding inclusion for B. Taking their union proves the claim.

(b) Show strict inclusion can occur.

Let $A = \{(0,0)\}$ and $B = \{(2,0)\}$. The union of their hulls has just two points, whereas the combined hull is the whole segment and includes $(1,0)$.

Exercise 8.3.20: Intersection of hulls can be larger

(a) Apply monotonicity twice.

Since $A \cap B \subseteq A$ and $A \cap B \subseteq B$, its hull is contained in each hull, hence in their intersection.

(b) Give an example with a new intersection point.

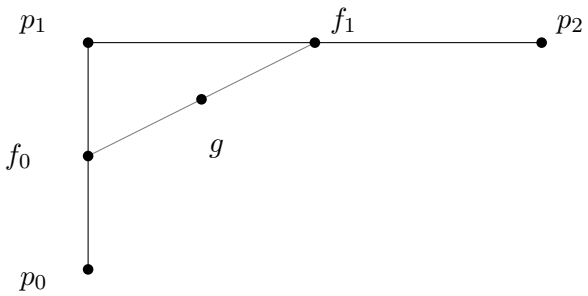
Take $A = \{(-1,0),(1,0)\}$ and $B = \{(0,-1),(0,1)\}$. The sets have empty intersection, so its hull is empty. But their two segments intersect at zero, showing that the claimed inclusion can be strict.

Exercise 8.3.21: The midpoint construction for a quadratic curve

At $t = 1/2$,

$$f_0 = \tfrac{1}{2}(p_0 + p_1), \quad f_1 = \tfrac{1}{2}(p_1 + p_2), \quad g = \tfrac{1}{2}(f_0 + f_1) = \tfrac{1}{4}p_0 + \tfrac{1}{2}p_1 + \tfrac{1}{4}p_2.$$

Thus first mark the midpoints of p_0p_1 and p_1p_2, join them, and mark the midpoint of that new segment. The original diagram below uses the same right-angle configuration as the source, with convenient schematic coordinates $p_0 = (0,0), p_1 = (0,2), p_2 = (4,2)$.

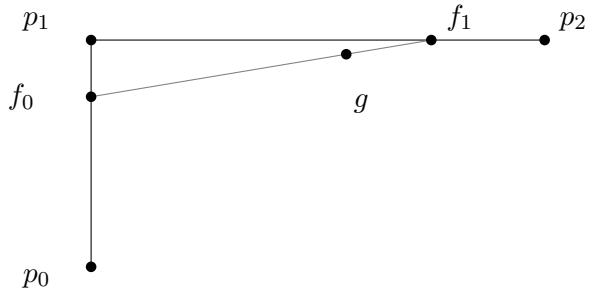


Exercise 8.3.22: A three-quarter construction

Now $f_0 = (1/4)p_0 + (3/4)p_1$ and $f_1 = (1/4)p_1 + (3/4)p_2$. Place each point three quarters of the way along its directed segment. Then place g three quarters of the way from f_0 to f_1:

$$g = \tfrac{1}{16}p_0 + \tfrac{3}{8}p_1 + \tfrac{9}{16}p_2.$$

The same schematic coordinates give $f_0 = (0, 3/2)$, $f_1 = (3, 2)$, and $g = (9/4, 15/8)$, as shown.



Exercise 8.3.23: Expand the quadratic Bézier formula

Substitute both definitions into $g = (1 - t)f_0 + tf_1$ and distribute:

$$g(t) = (1 - t)^2 p_0 + 2t(1 - t)p_1 + t^2 p_2.$$

For $0 \le t \le 1$, the three coefficients are nonnegative and their sum is $(1 - t)^2 + 2t(1 - t) + t^2 = ((1 - t) + t)^2 = 1$. Thus this formula is a convex combination of the control points and lies in their convex hull for the entire parameter interval.

Exercise 8.3.24: The cubic curve also stays inside its controls

Apply the preceding formula to the two quadratic curves and collect terms:

$$h(t) = (1 - t)^3 p_0 + 3t(1 - t)^2 p_1 + 3t^2(1 - t)p_2 + t^3 p_3.$$

The four coefficients are nonnegative on $[0, 1]$ and add to $((1 - t) + t)^3 = 1$ by the binomial theorem. Hence every point of the cubic curve is a convex combination of the four controls and lies in their convex hull. Equivalently, both quadratic points already lie in that convex hull, which is itself convex.

8.4 Hyperplanes

A linear functional has scalar output: $f : \mathbb{R}^n \to \mathbb{R}$. It can be written $f(x) = n^T x$ for a fixed vector n. When $n \neq 0$, the level set $[f : d] = \{x : n^T x = d\}$ is a hyperplane. Its two open sides have $f(x) < d$ and $f(x) > d$. To separate sets weakly, put them in opposite closed sides; to separate strictly, put them in opposite open sides. Here "open" and "closed" refer to the ambient $\mathbb{R}^n$, not just the set's own line or plane. In Euclidean space compact means closed and bounded.

Practice problem 8.4.P1: An explicit line where two planes intersect

The normal equations are $x_1 + x_2 - 2x_3 = -3$ and $-2x_1 + x_2 + 3x_3 = 7$, using $n_1 \cdot p_1 = -3$ and $n_2 \cdot p_2 = 7$. Subtracting the second equation from the first after elimination gives the reduced equations $x_1 - (5/3)x_3 = -10/3$, $x_2 - (1/3)x_3 = 1/3$. Thus their intersection is

$$x = (-10/3, 1/3, 0) + t(5/3, 1/3, 1), \qquad t \in \mathbb{R}.$$

Substitution verifies both plane equations for every t, and the single free variable shows that every point in the intersection has this form.

Exercise 8.4.1: An implicit equation for a line

The direction from $(-1, 4)$ to $(3, 1)$ is $(4, -3)$. Choose the perpendicular normal $n = (3, 4)$, since $3(4) + 4(-3) = 0$. Then $d = n \cdot (-1, 4) = -3 + 16 = 13$, so take $f(x, y) = 3x + 4y$ and $d = 13$. Both points satisfy this equation, and its nonzero normal makes it the unique line through those two distinct points.

Exercise 8.4.2: A second line functional

The direction between $(1, 4)$ and $(-2, -1)$ is $(-3, -5)$. A perpendicular normal is $(5, -3)$; its dot product with the direction is $-15 + 15 = 0$. At the first point it has value $5 - 12 = -7$. Thus $f(x, y) = 5x - 3y$ and $d = -7$ give the required line $[f : d]$.

Exercise 8.4.3: Open, closed, or neither

(a) $y > 0$ is **open**, not closed. Every point has a ball staying above the axis; $(0, 1/k)$ tends to the excluded origin.

(b) The segment $x = 2$, $1 \leq y \leq 3$ is **closed**, not open. It includes its endpoints and is the intersection of closed constraints. Every disk about a point on it contains points with $x \neq 2$.

(c) The segment $x = 2$, $1 < y < 3$ is **neither**. It has no ambient open disk, and $(2, 1 + 1/k)$ tends to an excluded endpoint.

(d) The positive branch $xy = 1$, $x > 0$ is **closed**, not open. For a convergent sequence on it, the limit satisfies $xy = 1$ by continuity; its x cannot be zero or negative, so the limit stays on this branch. No disk about a curve point lies entirely on the curve.

(e) The region $xy \geq 1$, $x > 0$ is **closed**, not open. For a finite convergent sequence its limit has $x \geq 0$ and $xy \geq 1$, so $x = 0$ is impossible and the limit remains in the set. Boundary points with $xy = 1$ prevent openness.

Exercise 8.4.4: Five curved planar sets

(a) The circle $x^2 + y^2 = 1$ is **closed**, not open, by continuity of $x^2 + y^2$ and the absence of any disk contained in a circle.

(b) The region $x^2 + y^2 > 1$ is **open**, not closed: continuity preserves its strict inequality locally, but $(1 + 1/k, 0)$ tends to the excluded circle.

(c) The region $x^2 + y^2 \leq 1$, $y > 0$ is **neither**. The included arc point $(0, 1)$ is not interior; $(0, 1/k)$ tends to the excluded origin on the diameter.

(d) The region $y \geq x^2$ is **closed**, not open. It is defined by the nonnegative value of the continuous function $y - x^2$, and equality points have nearby points below the boundary.

(e) The region $y < x^2$ is **open**, not closed. It is a strict continuous inequality; for example $(0, -1/k)$ tends to the excluded origin.

Exercise 8.4.5: Compactness and convexity for the sets of Exercise 3

(a) The upper open half-plane is unbounded, hence not compact, but it is convex because averages of positive y-coordinates remain positive.

(b) The closed vertical segment is closed and bounded, hence compact. A segment is convex.

(c) The open vertical segment is not closed, hence not compact. It is convex: averaging two coordinates strictly between 1 and 3 keeps that coordinate strictly between the same bounds.

(d) The positive reciprocal curve is unbounded, hence not compact. It is not convex: the midpoint of $(1, 1)$ and $(2, 1/2)$ is $(3/2, 3/4)$, whose coordinate product is $9/8 \neq 1$.

(e) The region on and above the reciprocal curve is unbounded, hence not compact. It is convex: for positive x_1, x_2 and $0 \leq t \leq 1$,

$$\frac{1}{(1-t)x_1 + tx_2} \leq \frac{1-t}{x_1} + \frac{t}{x_2}.$$

This reciprocal convexity inequality, together with $y_i \geq 1/x_i$, puts each segment point on or above the same curve. Its horizontal coordinate stays positive. The inequality also follows by expanding the nonnegative quantity $t(1-t)(x_1 - x_2)^2$ after clearing positive denominators.

Exercise 8.4.6: Compactness and convexity for the sets of Exercise 4

(a) The circle is compact (closed and bounded) but not convex: $(1, 0)$ and $(-1, 0)$ have midpoint zero, off the circle.

(b) The exterior of the disk is unbounded and not compact. It is not convex: $(2, 0), (-2, 0)$ are in it but their midpoint is not.

(c) The upper half-disk with the diameter excluded is not closed, hence not compact. It is convex as the intersection of the closed unit disk and the convex open half-plane $y > 0$.

(d) The epigraph $y \geq x^2$ is unbounded, hence not compact. It is convex because $((1-t)x_1 + tx_2)^2 \leq (1-t)x_1^2 + tx_2^2 \leq (1-t)y_1 + ty_2$. The first inequality follows by subtracting the left side to get $t(1-t)(x_1 - x_2)^2 \geq 0$.

(e) The region $y < x^2$ is unbounded and not compact. It is not convex: $(-1, 1/2), (1, 1/2)$ are in it, but their midpoint $(0, 1/2)$ is not.

Exercise 8.4.7: Find the normal from difference vectors

Subtract the first point from every other listed point to get directions parallel to the hyperplane. Solve $D^T n = 0$, where these directions form the columns of D. One nonzero solution is

$$n = (0, 2, 3)^T, \qquad n^T p_1 = 11.$$

Checking gives $n^T(p_j - p_1) = 0$ for every other listed point. Thus (a) this n is a normal and (b) the required functional is $f(x) = n^T x$, with $d = 11$. The equation $n^T x = 11$ contains all the points. The difference vectors have rank 2, so their affine hull has the correct hyperplane dimension; the equation describes that whole hyperplane.

Exercise 8.4.8: Find the normal from difference vectors

Subtract the first point from every other listed point to get directions parallel to the hyperplane. Solve $D^T n = 0$, where these directions form the columns of D. One nonzero solution is

$$n = (-4, -3, 6)^T, \qquad n^T p_1 = 8.$$

Checking gives $n^T(p_j - p_1) = 0$ for every other listed point. Thus (a) this n is a normal and (b) the required functional is $f(x) = n^T x$, with $d = 8$. The equation $n^T x = 8$ contains all the points. The difference vectors have rank 2, so their affine hull has the correct hyperplane dimension; the equation describes that whole hyperplane.

Exercise 8.4.9: Find the normal from difference vectors

Subtract the first point from every other listed point to get directions parallel to the hyperplane. Solve $D^T n = 0$, where these directions form the columns of D. One nonzero solution is

$$n = (3, -1, 2, 1)^T, \qquad n^T p_1 = 5.$$

Checking gives $n^T(p_j - p_1) = 0$ for every other listed point. Thus (a) this n is a normal and (b) the required functional is $f(x) = n^T x$, with $d = 5$. The equation $n^T x = 5$ contains all the points. The difference vectors have rank 3, so their affine hull has the correct hyperplane dimension; the equation describes that whole hyperplane.

Exercise 8.4.10: Find the normal from difference vectors

Subtract the first point from every other listed point to get directions parallel to the hyperplane. Solve $D^T n = 0$, where these directions form the columns of D. One nonzero solution is

$$n = (-2, 3, -5, 1)^T, \qquad n^T p_1 = 4.$$

Checking gives $n^T(p_j - p_1) = 0$ for every other listed point. Thus (a) this n is a normal and (b) the required functional is $f(x) = n^T x$, with $d = 4$. The equation $n^T x = 4$ contains all the points. The difference vectors have rank 3, so their affine hull has the correct hyperplane dimension; the equation describes that whole hyperplane.

Exercise 8.4.11: Use the sign of the functional relative to its level

Here $f(x) = 2x_1 + x_2 + 5x_3 - x_4$ and $d = f(p) = 2 - 3 + 5 - 2 = 2$. At the origin $f(0) - d = -2 < 0$. For the three proposed points the functional values are $f(v_1) = 5$, $f(v_2) = -2$, and $f(v_3) = 2$. Thus v_2 is on the same open side as the origin, v_1 is on the opposite side, and v_3 lies in the hyperplane itself, on neither open side.

Exercise 8.4.12: Weak separation without strict parallel separation

With the prescribed normal $n = (3, 1, -2)$, use $f(x) = 3x_1 + x_2 - 2x_3$. Evaluation at the three points of A gives $-5, 4, 3$; at those of B it gives $7, 4, 6$. Hence $H = [f : 4]$ separates the sets weakly: $f(A) \le 4 \le f(B)$. But $\max f(A) = 4 = \min f(B)$, so there is no real number strictly between these two bounds. No hyperplane parallel to H strictly separates them. This only answers the prescribed-normal question; it does not assert that no differently oriented separating hyperplane could exist.

Exercise 8.4.13: The intersection of two hyperplanes in four dimensions

The two constants are $n_1 \cdot p_1 = 4$ and $n_2 \cdot p_2 = 22$. Thus solve

$$x_1 + 2x_2 + 4x_3 + 2x_4 = 4, \qquad 2x_1 + 3x_2 + x_3 + 5x_4 = 22.$$

Elimination gives $x_1 - 10x_3 + 4x_4 = 32$ and $x_2 + 7x_3 - x_4 = -14$. Take $x_3 = s, x_4 = t$ freely. The explicit answer is

$$x = (32, -14, 0, 0) + s(10, -7, 1, 0) + t(-4, 1, 0, 1), \qquad s, t \in \mathbb{R}.$$

Substitution checks both equations. The two direction vectors are independent because their third and fourth coordinates isolate s, t. Every solution was obtained through the two free variables, so this is the entire intersection, a two-dimensional flat.

Exercise 8.4.14: Possible dimensions of an intersection of flats

Choose $p \in F_1 \cap F_2$ and translate it to zero. Then $F_1 = p + U$, $F_2 = p + V$ with $\dim U = \dim V = 4$, and $F_1 \cap F_2 = p + (U \cap V)$. The dimension formula gives

$$\dim(U \cap V) = \dim U + \dim V - \dim(U + V) \ge 8 - 6 = 2,$$

and the intersection has dimension at most four. Each possibility occurs: take $U = \text{Span}\{e_1, e_2, e_3, e_4\}$ in $\mathbb{R}^6$, and take V respectively as $\text{Span}\{e_1, e_2, e_5, e_6\}$, $\text{Span}\{e_1, e_2, e_3, e_5\}$, or U. The intersection dimensions are $2, 3, 4$.

Exercise 8.4.15: An already implicit hyperplane equation

The equation $Ax = b$ is $x_1 - 3x_2 + 4x_3 - 2x_4 = 5$. Thus $f(x) = x_1 - 3x_2 + 4x_3 - 2x_4$ and $d = 5$. The coefficient vector is nonzero, so its level set is a hyperplane in $\mathbb{R}^4$.

Exercise 8.4.16: A null space of codimension one

Set $f(x) = 2x_1 + 5x_2 - 3x_3 + 6x_5$ and $d = 0$. Then $[f : 0]$ is exactly the given null space in $\mathbb{R}^5$. The fourth coordinate is free because its coefficient is zero; the functional itself is nonzero, so this is a hyperplane rather than the whole space.

Exercise 8.4.17: Describe a row plane by its perpendicular direction

Solve $Bn = 0$ for $B = \left(\begin{smallmatrix} 1 & 3 & 5 \\ 0 & 2 & 4 \end{smallmatrix}\right)$. The second equation gives $n_2 = -2n_3$, the first then gives $n_1 = n_3$. Take $n = (1, -2, 1)$. Both independent rows are perpendicular to n, so their two-dimensional span equals $\{x : n^T x = 0\}$. Thus $f(x) = x_1 - 2x_2 + x_3$, $d = 0$.

Exercise 8.4.18: A second row plane

For $B = \left(\begin{smallmatrix} 1 & 4 & -5 \\ 0 & -2 & 8 \end{smallmatrix}\right)$, solving $Bn = 0$ gives $n_2 = 4n_3$ and $n_1 = -11n_3$. Choose $n = (-11, 4, 1)$. Since the two rows of B are independent, their span is the plane $-11x_1 + 4x_2 + x_3 = 0$. Take $f(x) = -11x_1 + 4x_2 + x_3$ and $d = 0$.

Exercise 8.4.19: A column plane and a transpose null space

A normal must be perpendicular to both columns, so solve $B^T n = 0$: $n_1 + 4n_2 - 7n_3 = 0$, $2n_2 - 6n_3 = 0$. These give $n_2 = 3n_3$, $n_1 = -5n_3$. Thus $n = (-5, 3, 1)$ is a normal. The columns are independent, so the column space is the full plane $-5x_1 + 3x_2 + x_3 = 0$. Take this expression as f and take $d = 0$.

Exercise 8.4.20: A second column-plane equation

The normal equations are $n_1 + 5n_2 - 4n_3 = 0$ and $2n_2 - 4n_3 = 0$. Choose $n_3 = 1$, giving $n = (-6, 2, 1)$. The two original columns are independent, so $H = \text{Col}\, B$ equals $\{x : -6x_1 + 2x_2 + x_3 = 0\}$. Hence $f(x) = -6x_1 + 2x_2 + x_3$, $d = 0$.

Exercise 8.4.21: Four assertions about functionals and separation

(a) **False.** A linear functional maps $\mathbb{R}^n$ *to* $\mathbb{R}$, not generally from $\mathbb{R}$ to $\mathbb{R}^n$.

(b) **False.** The correct general formula is $f(x) = n^T x$, a dot product with a fixed vector, not scalar multiplication kx, which has vector output when $n > 1$. For example $f(x_1, x_2) = x_1 + x_2$ is a functional.

(c) **True.** A point in both sets would have its functional value simultaneously strictly below and strictly above the same level d. That is impossible.

(d) **False.** Two disjoint closed convex sets need not have a positive separating gap if neither is compact. For example $A = \{(x, y) : y \geq e^x\}$ and $B = \{(x, 0) : x \in \mathbb{R}\}$ are closed, convex, and disjoint. Any functional bounded on one side along the entire line B must have zero x-coefficient. Separation would then have to be horizontal, but A has positive y values arbitrarily close to zero, while B has $y = 0$. No horizontal level lies strictly between the sets.

Exercise 8.4.22: Four further assertions about separating hyperplanes

(a) **True.** A nonzero functional has a nonzero coefficient vector, so each of its real level sets is a hyperplane.

(b) **False.** The normal must be nonzero. If $n = 0, d = 0$, the set is all of $\mathbb{R}^n$; if $n = 0, d \neq 0$, it is empty. Neither is a hyperplane.

(c) **False.** Convexity is missing. Take $A = \{0\}$ and B the unit circle. They are disjoint, with A compact and B closed, but the center cannot be strictly separated from all points of the surrounding circle. If $b \in B$, both b and $-b$ belong to it, so no functional can put both strictly on the same side of a level separating them from zero.

(d) **False.** Failure of one chosen hyperplane does not imply failure of every choice. Take $A = \{(0, 0)\}$, $B = \{(2, 0)\}$, and the line $y = 100$, which does not strictly separate them. Their convex hulls are still disjoint, and the line $x = 1$ does strictly separate them.

Exercise 8.4.23: Separate a point from a triangle by an edge normal

Take $f(x, y) = 3x - 2y$. Its values at the three listed triangle vertices are $1, 9, 9$, whereas $f(p) = 10$ for $p = (4, 1)$. Every convex combination of the vertices therefore has value at most 9. Thus the line $[f : 19/2]$, or $3x - 2y = 9.5$, strictly separates the triangle from p: all its hull values are below 9.5, and the point's value is above.

Exercise 8.4.24: A second point outside a triangle

Take $f(x, y) = -2x + 3y$. At $(1, 2), (5, 1), (4, 4)$ the values are $4, -7, 4$; at $p = (2, 3)$ the value is 5. Thus $[f : 9/2]$ strictly separates p from the triangle, since its entire convex hull has functional value at most $4 < 4.5 < 5$.

Exercise 8.4.25: Separate two open balls using their center direction

Let $p = (4, 1)$, $f(x) = p \cdot x = 4x_1 + x_2$, and $v = (3/4)p = (3, 3/4)$. Set $d = f(v) = 51/4$. For $x \in B(0, 3)$, Cauchy–Schwarz gives $f(x) < 3\sqrt{17} < 51/4$. For $x \in B(p, 1)$, it gives $f(x) = 17 + p \cdot (x - p) > 17 - \sqrt{17} > 51/4$. Thus $[f : 51/4]$ strictly separates the balls. The hinted point v lies in neither

ball because $\|v\| = 3\sqrt{17}/4 > 3$ and $\|v - p\| = \sqrt{17}/4 > 1$. The strict inequalities are justified by $\sqrt{17} < 17/4$ and $3\sqrt{17} < 51/4$, which follow on squaring positive quantities.

Exercise 8.4.26: Translate the two-ball construction

The center difference is $n = p - q = (4, -2)$, with $\|n\| = \sqrt{20}$. Take $f(x) = 4x_1 - 2x_2$, so $f(q) = 2$, $f(p) = 22$. The point $v = q + (3/4)(p - q) = (5, 3/2)$ has $f(v) = 17$. For $x \in B(q, 3)$, $f(x) < 2 + 3\sqrt{20} < 17$; for $x \in B(p, 1)$, $f(x) > 22 - \sqrt{20} > 17$. Therefore $[f : 17]$ strictly separates the balls. The inequalities follow from $\sqrt{20} < 5$, and they leave a positive gap between the sets.

Exercise 8.4.27: A closed set whose convex hull loses closedness

Let

$$S = \{(n, 1/n), (-n, 1/n) : n = 1, 2, 3, \ldots\}.$$

This set is closed: any convergent sequence in it is bounded, so its integer first coordinates range through only finitely many possibilities; its limit is in that finite subset of S. But the midpoint of $(n, 1/n)$ and $(-n, 1/n)$ is $(0, 1/n)$, so these points lie in $\operatorname{conv} S$ and tend to $(0, 0)$. No convex combination of points in S has height zero, because it is a finite weighted average of strictly positive heights. Thus $(0, 0)$ is not in the convex hull, which is consequently not closed.

Exercise 8.4.28: Disjoint hulls need not give a strict separating gap

Take the closed set $B = S$ of Exercise 8.4.27 and the compact set $A = \{(0, 0)\}$. Every point in $\operatorname{conv} B$ has positive height, so the two convex hulls are disjoint. Suppose a hyperplane strictly separated A and B. After changing the functional's sign if necessary, its level would satisfy $0 = f(0) < d < f(b)$ for every $b \in B$. Taking the midpoint of the two points with first coordinates $\pm n$ would give $f(0, 1/n) > d$. Letting $n \to \infty$ contradicts $d > 0$, since $f(0, 1/n) \to 0$. Thus strict separation is impossible. The compact-set separation theorem cannot be applied to $\operatorname{conv} B$ as a closed set here: Exercise 8.4.27 shows that it is not closed.

Exercise 8.4.29: An open ball is convex

For $x, y \in B(p, \delta)$ and $0 \le t \le 1$, the triangle inequality gives

$$\|(1 - t)x + ty - p\| = \|(1 - t)(x - p) + t(y - p)\| \le (1 - t)\|x - p\| + t\|y - p\| < \delta.$$

At $t = 0$ or 1 this is the original strict inequality; between them it is a weighted average of two numbers strictly below δ. Thus the whole segment remains inside the open ball.

Exercise 8.4.30: Convex hull preserves boundedness

Since S is bounded, choose R with $\|s\| \le R$ for every $s \in S$. For a convex combination $y = \sum_i \lambda_i s_i$, use the triangle inequality, nonnegativity, and sum-one property to obtain

$$\|y\| \le \sum_i \lambda_i \|s_i\| \le \sum_i \lambda_i R = R.$$

Hence the same ball bounds $\operatorname{conv} S$. If S is empty its hull is empty and bounded as well.

8.5 Polytopes

A polytope is the convex hull of finitely many points. Its minimal representation consists exactly of its extreme points, or vertices. For a linear functional, the value at a convex combination is the same combination of the values at its generators. Hence its maximum is the largest generator value; the full maximizing set is the convex hull of *all* generators tying for that largest value. The analogous statement holds for minima. A k-face has dimension k; vertices have dimension zero, edges dimension one, and facets of an n-polytope dimension $n-1$.

Practice problem 8.5.P1: Turn five inequalities into a vertex list

In the first quadrant the constraints are $x+3y\le 12$, $x+2y\le 9$, $2x+y\le 12$. The axis vertices are $(0,0),(6,0),(0,4)$, obtained by imposing the most restrictive intercept on each axis. Pairwise boundary intersections are $(3,3),(5,2),(24/5,12/5)$. The last violates $x+2y\le 9$, because $24/5+24/5=48/5>9$. The first two pass the remaining inequalities. Thus the minimal representation is

$$P=\operatorname{conv}\{(0,0),(6,0),(5,2),(3,3),(0,4)\}.$$

Each retained point is a genuine corner of the feasible polygon; no retained point is a convex combination of the others. The printed final sentence calls these "three" vertices while listing five; there are five.

Exercise 8.5.1: All maximizers of three functionals on a triangle

At $p_1=(1,0), p_2=(2,3), p_3=(-1,2)$, the functional values are

	p_1	p_2	p_3	max	all maximizing points
$x-y$	1	-1	-3	1	p_1
$x+y$	1	5	1	5	p_2
$-3x+y$	-3	-3	5	5	p_3

These respectively answer (a), (b), (c). To justify completeness, write $x=\sum_i \lambda_i p_i$ with nonnegative coefficients of sum one. If a positive coefficient were attached to a vertex with value below the maximum, their weighted average would be strictly below that maximum. Here each maximum occurs at just one vertex, so no other point can attain it.

Exercise 8.5.2: Two maximizing edges

For $p_1=(0,-1), p_2=(2,1), p_3=(1,2)$, evaluation gives

	p_1	p_2	p_3	max	all maximizing points
$x+y$	-1	3	3	3	$\operatorname{conv}\{p_2,p_3\}$
$x-y$	1	1	-1	1	$\operatorname{conv}\{p_1,p_2\}$
$-2x+y$	-1	-3	0	0	p_3

In each tied case every point on the listed edge has the same value by linearity. A convex combination involving the lower-value vertex with positive weight has smaller value, so the whole maximizing set is exactly that edge.

Exercise 8.5.3: Minima for the first triangle

Use the values from Exercise 8.5.1 and select the smallest in each row. For (a), the minimum of $x - y$ is -3, attained only at p_3. For (b), the minimum of $x + y$ is 1, attained at every point of the edge $\text{conv}\{p_1, p_3\}$ and nowhere else. For (c), the minimum of $-3x + y$ is -3, attained exactly on $\text{conv}\{p_1, p_2\}$. The weighted-average argument used for maxima also proves these sets contain all minimizers.

Exercise 8.5.4: Minima for the second triangle

From the values in Exercise 8.5.2: (a) $x + y$ has minimum -1 only at p_1; (b) $x - y$ has minimum -1 only at p_3; (c) $-2x + y$ has minimum -3 only at p_2. No other point can attain one of these strict vertex minima, because a combination using another vertex would increase the weighted average.

Exercise 8.5.5: A polygon bounded by two sloping lines

The inequalities are $x + 2y \leq 10$, $3x + y \leq 15$, $x, y \geq 0$. The axis endpoints are $(5, 0)$ and $(0, 5)$. Solve both boundary equations: substitute $y = 15 - 3x$ into the first, obtaining $x + 30 - 6x = 10$, so $x = 4, y = 3$. This intersection is in the first quadrant. Together with the origin these are all corners:

$$P = \text{conv}\{(0, 0), (5, 0), (4, 3), (0, 5)\}.$$

Each corner is the meeting point of two boundary lines of the feasible polygon, and each lies on its boundary with a change in direction, so none is redundant in the minimal representation.

Exercise 8.5.6: Another two-constraint polygon

Here $2x + 3y \leq 18$, $4x + y \leq 16$, $x, y \geq 0$. The axis endpoints are $(4, 0), (0, 6)$. The two sloping boundaries intersect where $y = 16 - 4x$ and $2x + 3(16 - 4x) = 18$, giving $x = 3, y = 4$. Thus

$$P = \text{conv}\{(0, 0), (4, 0), (3, 4), (0, 6)\}.$$

All listed points satisfy all constraints and are the successive corners of the feasible polygon, hence give its minimal representation.

Exercise 8.5.7: Three boundaries meet at one corner

The constraints are $x + 3y \leq 18$, $x + y \leq 10$, $4x + y \leq 28$, with nonnegative coordinates. The restrictive axis intercepts are $(7, 0)$ and $(0, 6)$. Solving any pair of the sloping equalities gives the same point $(6, 4)$, which satisfies all three with equality. Hence there is only one non-axis corner, not three:

$$P = \text{conv}\{(0, 0), (7, 0), (6, 4), (0, 6)\}.$$

The inequality $x + y \leq 10$ touches this corner but introduces no new vertex.

Exercise 8.5.8: A redundant inequality

The constraints are $2x + y \leq 8$, $x + y \leq 6$, $x + 2y \leq 7$, with $x, y \geq 0$. Their restrictive axis endpoints are $(4, 0), (0, 7/2)$. The first and third boundaries meet at $(3, 2)$, which satisfies $x + y \leq 6$. The intersections with the middle boundary are $(2, 4)$ and $(5, 1)$; they violate respectively $x + 2y \leq 7$ and $2x + y \leq 8$ and must be discarded. Therefore

$$P = \text{conv}\{(0, 0), (4, 0), (3, 2), (0, 7/2)\}.$$

The discarded points are intersections of boundary lines, but they are not feasible and therefore cannot be vertices of this polytope.

Exercise 8.5.9: An extreme point that is not an exposed vertex

Let $D = \{(x, y) : x^2 + (y - 1)^2 \leq 1\}$ and $K = \text{conv}(D \cup \{(3, 0)\})$. Every point of K has $y \geq 0$, and its points with $y = 0$ form precisely the segment $[(0, 0), (3, 0)]$. Indeed the disk itself touches $y = 0$ only at the origin. If the origin were a strict convex combination of two distinct points in K, their nonnegative heights would both have to be zero. Their nonnegative horizontal coordinates on that segment could then average to zero only if both points were the origin. Thus the origin is extreme. However, the only supporting line to the disk at the origin is its tangent $y = 0$. Any supporting line to K there must also support the contained disk, so it must be that same line. It intersects K in the whole segment to $(3, 0)$, not in a singleton.

Conclusion. The origin is extreme but is not a vertex in the section's supporting-hyperplane (zero-dimensional exposed-face) sense.

Exercise 8.5.10: A closed unbounded counterexample to profile generation

Take the ray $S = \{(x, 0) : x \geq 0\}$. It is closed and convex. Its origin is extreme, while every point with $x > 0$ is the midpoint of two other ray points. Thus its profile is $\{(0, 0)\}$. The convex hull of that profile is just the singleton, not the whole ray. This shows why closedness without boundedness is insufficient.

Exercise 8.5.11: A bounded but nonclosed counterexample

Take the half-open segment $S = \{(x, 0) : 0 \leq x < 1\}$. It is bounded and convex. Its profile is again just the origin, since any other point has two distinct nearby points on either side inside the segment. The hull of the nonempty profile is a singleton, not S. This time the missing hypothesis is closedness.

Exercise 8.5.12: Face counts for simplices

(a) Count the faces of the five-simplex.

A five-simplex has six vertices; each k-face is determined by choosing $k + 1$ of them. Thus its counts for $k = 0, 1, 2, 3, 4$ are $6, 15, 20, 15, 6$. Their alternating sum is $6 - 15 + 20 - 15 + 6 = 2$, equal to $1 + (-1)^4$, as Euler's formula requires.

(b) Display the pattern and explain it.

$n\backslash k$	0	1	2	3	4
1	2	–	–	–	–
2	3	3	–	–	–
3	4	6	4	–	–
4	5	10	10	5	–
5	6	15	20	15	6

The general formula is $f_k(S^n) = \binom{n+1}{k+1}$ for $0 \le k \le n-1$. Affine independence ensures that each chosen $k+1$ vertices defines a k-simplex face and that different subsets give different faces. Dashes denote dimensions not belonging to the list of proper faces.

Exercise 8.5.13: Face counts for hypercubes

(a) Count a five-cube and check Euler.

Choose k coordinates allowed to vary and fix each remaining coordinate at either endpoint. Thus $f_k(C^n) = 2^{n-k}\binom{n}{k}$. For $n = 5$ the counts are $32, 80, 80, 40, 10$; $32 - 80 + 80 - 40 + 10 = 2 = 1 + (-1)^4$.

(b) The requested chart.

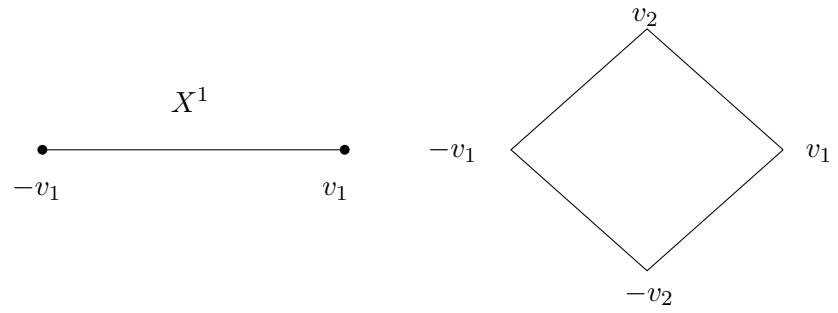
$n\backslash k$	0	1	2	3	4
1	2	–	–	–	–
2	4	4	–	–	–
3	8	12	6	–	–
4	16	32	24	8	–
5	32	80	80	40	10

The coordinate-choice argument proves the formula, rather than only predicting it from these few values.

Exercise 8.5.14: Crosspolytopes in increasing dimensions

(a) Draw the first two cases.

X^1 is the segment joining $-v_1$ to v_1. X^2 is the parallelogram with successive vertices $v_1, v_2, -v_1, -v_2$. Below are representatives using standard-coordinate generators; an invertible linear change gives the analogous pictures for any independent generators.



(b) The three-dimensional case.

There are 6 vertices, 12 edges, and 8 triangular facets. The standard-coordinate version is a regular octahedron; for arbitrary independent generators it is a linearly distorted octahedron with the same face counts.

(c) The four-dimensional case.

Its counts are $8, 24, 32, 16$. The alternating sum is $8 - 24 + 32 - 16 = 0 = 1 + (-1)^3$, as required.

(d) Explain the general formula.

A proper face selects $k+1$ of the n opposite vertex pairs, and selects one sign from each selected pair; both opposite vertices cannot belong to one proper face. These choices give

$$f_k(X^n) = 2^{k+1}\binom{n}{k+1}, \qquad 0 \le k \le n-1.$$

Every such selection is a simplex face: for the standard crosspolytope a supporting functional assigning the selected signed coordinates value one and all other coordinate coefficients zero exposes exactly that face. Linear equivalence preserves the face structure for general independent generators.

Exercise 8.5.15: Face counts for a pyramid

Let the $(n-1)$-dimensional base be Q and the apex lie outside its affine hull.

(a) Vertices.

All base vertices remain vertices and the apex adds one, so $f_0(P^n) = f_0(Q) + 1$.

(b) Intermediate dimensions.

A k-face not containing the apex is a k-face of Q. A k-face containing the apex is the pyramid over a $(k-1)$-face of Q. Thus $f_k(P^n) = f_k(Q) + f_{k-1}(Q)$ for $1 \le k \le n-2$.

(c) Facets.

There is one base facet Q; each of its $(n-2)$-faces joined to the apex gives a side facet. Therefore $f_{n-1}(P^n) = 1 + f_{n-2}(Q)$, for $n \ge 2$. These types are disjoint and cover every face of a pyramid.

Exercise 8.5.16: Four statements about profiles and simplices

(a) **True.** This is the definition of a polytope.

(b) **True.** An extreme point cannot lie strictly between two other points of the convex set. If $p \in uv$ and $p \neq u$, it must equal v.

(c) **False.** Compactness is necessary in the profile theorem; the ray in Exercise 8.5.10 is a counterexample.

(d) **True.** A four-simplex has five vertices; omit one at a time to obtain its five facets, each a three-simplex, or tetrahedron.

Exercise 8.5.17: Four more statements about faces

(a) **False.** A three-dimensional cube has six square facets.

(b) **True.** For polytopes the minimal-representation theorem identifies extreme points with vertices. The distinction seen for a nonpolytope in Exercise 8.5.9 does not occur for polytopes.

(c) **False.** A maximizing point need not itself be extreme. On the square $[-1,1]^2$, $f(x,y)=x$ is maximal at $(1,0)$, the midpoint of the edge joining $(1,-1)$ and $(1,1)$.

(d) **True.** A two-dimensional polytope is a polygon; each edge joins successive vertices around its boundary, giving equally many vertices and edges.

Exercise 8.5.18: Removing an extreme point preserves convexity

Suppose v is extreme and $x,y \in S \setminus \{v\}$. Their segment lies in S by convexity and cannot pass through v in its interior, by extremality; neither endpoint is v. Thus the whole segment lies in $S \setminus \{v\}$, proving that set convex. Conversely, if $S \setminus \{v\}$ is convex and v were strictly between some distinct $x,y \in S$, neither could equal v. Their segment would lie in the set with v removed, contradicting its containing v. Hence v is extreme. Singleton cases work because the empty set is convex.

Exercise 8.5.19: Add two positive dilations of one convex set

An arbitrary element of $cS+dS$ is $cx+dy$ with $x,y \in S$. Since $c,d>0$,

$$cx+dy=(c+d)\left(\frac{c}{c+d}x+\frac{d}{c+d}y\right)\in(c+d)S$$

by convexity. This proves one inclusion. Conversely, for each $s \in S$, $(c+d)s=cs+ds \in cS+dS$, proving the reverse inclusion. Thus $cS+dS=(c+d)S$.

Exercise 8.5.20: Convexity cannot be omitted

Take $S=\{(0,0),(1,0)\}$ and $c=d=1$. Then $S+S=\{(0,0),(1,0),(2,0)\}$, whereas $2S=\{(0,0),(2,0)\}$. The point $(1,0)$ occurs only in the sum. Thus the equality from Exercise 8.5.19 can fail for a nonconvex set.

Exercise 8.5.21: A Minkowski sum of convex sets is convex

Take two points $x=a_1+b_1$ and $y=a_2+b_2$ in $A+B$. For $0 \le t \le 1$,

$$(1-t)x+ty=((1-t)a_1+ta_2)+((1-t)b_1+tb_2).$$

The first parenthesis belongs to A and the second to B by their convexity. Thus their sum is in $A+B$. This proves the entire segment is included, as required.

Exercise 8.5.22: Why the list of regular polyhedra has five entries

(a) Count incidences in two ways.

Every one of the r facets has k edges, giving kr facet-edge incidences. Each edge belongs to two facets, giving $2e$; hence $kr = 2e$. Likewise every one of the v vertices meets s edges, while every edge has two endpoints, so $sv = 2e$.

(b) Substitute those counts into Euler's formula.

From $v - e + r = 2$, substitute $v = 2e/s$ and $r = 2e/k$, then divide by $2e$:

$$\frac{1}{s} + \frac{1}{k} = \frac{1}{2} + \frac{1}{e}.$$

(c) Exhaust the possible integer pairs.

A face has at least three sides, so $k \geq 3$; at least three edges meet at any vertex, so $s \geq 3$. The equation makes $1/s + 1/k > 1/2$. If $k \geq 6$, its left side is at most $1/3 + 1/6 = 1/2$, impossible. For $k = 3$, the strict inequality gives $s < 6$, so $s = 3, 4, 5$. For $k = 4$, it gives $s < 4$, so $s = 3$. For $k = 5$, it gives $s < 10/3$, again $s = 3$. The counts follow from $e = (1/s + 1/k - 1/2)^{-1}$ and the incidence formulas:

k	s	v	e	r	polyhedron
3	3	4	6	4	tetrahedron
4	3	8	12	6	cube
3	4	6	12	8	octahedron
5	3	20	30	12	dodecahedron
3	5	12	30	20	icosahedron

These are the five realized regular convex polyhedra named in the exercise, and the integer argument rules out every other possibility.

8.6 Curves and surfaces

A cubic Bézier curve has four controls and formula

$$B(t) = (1-t)^3 p_0 + 3t(1-t)^2 p_1 + 3t^2(1-t)p_2 + t^3 p_3, \qquad 0 \le t \le 1.$$

The coefficients are nonnegative and sum to one, so the curve lies in the convex hull of its controls. A uniform cubic B-spline segment is a *different* curve, with weights

$$S(t) = \tfrac{1}{6}\big[(1-t)^3 p_0 + (3t^3 - 6t^2 + 4)p_1 + (-3t^3 + 3t^2 + 3t + 1)p_2 + t^3 p_3\big].$$

We use this formula, identical to the Practice Problems' equations (14)–(15), for the B-spline calculations below. Exercise 2 prints an inconsistent coefficient; its solution explicitly diagnoses the discrepancy before correcting it. The numbered exercises here concern curve segments and subdivision; the section's surface discussion uses these same one-variable operations in both parameter directions.

Practice problem 8.6.P1: The initial endpoint of a B-spline segment

Substitute $t = 0$ into $S(t)$ to obtain

$$S(0) = \tfrac{1}{6}p_0 + \tfrac{2}{3}p_1 + \tfrac{1}{6}p_2.$$

The coefficients are nonnegative and sum to one, so this point is in $\operatorname{conv}\{p_0, p_1, p_2\}$. Under the stated affine-independence assumption its unique affine coordinates are $(1/6, 2/3, 1/6)$. They differ from $(1, 0, 0)$, the coordinates of p_0, so it is not p_0 under that assumption. Without independence the formula still holds, but a degenerate choice such as all controls equal can make the endpoint coincide with a control.

Practice problem 8.6.P2: The final endpoint

At $t = 1$ the coefficients give

$$S(1) = \tfrac{1}{6}p_1 + \tfrac{2}{3}p_2 + \tfrac{1}{6}p_3.$$

Hence it is in $\operatorname{conv}\{p_1, p_2, p_3\}$. When those points are affinely independent, its affine coordinates are $(1/6, 2/3, 1/6)$, not $(0, 0, 1)$, so it differs from p_3. This proves the claimed endpoint behavior with its required nondegeneracy condition.

Exercise 8.6.1: Translate every control by the same vector

Write the four Bézier weights as $b_j(t)$. Their sum is $((1-t)+t)^3 = 1$. Therefore

$$B(t) + b = \sum_{j=0}^{3} b_j(t)p_j + \left(\sum_{j=0}^{3} b_j(t)\right) b = \sum_{j=0}^{3} b_j(t)(p_j + b).$$

This is a cubic Bézier curve with translated controls $p_j + b$. Translation is affine rather than linear, and the sum-one identity is exactly what makes the same control transformation work.

Exercise 8.6.2: Diagnose the printed coefficient before proving the B-spline properties

The exercise prints $3t(1-t) - 3t + 4$ as the coefficient of p_1 before the factor $1/6$, whereas the preceding Practice Problem equation (15) has $3t(1-t)^2 - 3t + 4$. The missing square changes the mathematics. With the printed unsquared expression, the four weights sum to

$$1 + \tfrac{1}{2}t^2(1-t),$$

not one. At $t = 1/2$ they sum to $17/16$. For example, if all four controls are one fixed nonzero vector p, the printed curve gives $(17/16)p$, which is outside their singleton convex hull. Thus part (a) is false literally as printed. We now use the consistent B-spline formula (14)–(15), with that square restored, for this exercise and later references to it.

(a) Prove convex-hull containment for the intended formula.

The first and fourth weights are $(1-t)^3/6$ and $t^3/6$, hence nonnegative. For the second use $3t(1-t)^2 - 3t + 4 \geq 4 - 3t \geq 1$ on $[0,1]$. For the third use $3t^2(1-t) + 3t + 1 \geq 1$. Expanding the four numerators makes their sum 6, so their weights sum to one. Thus $S(t)$ is a convex combination of the controls.

(b) Translate using the same sum-one property.

If the four weights are $s_j(t)$, then $S(t) + b = \sum_j s_j(t)(p_j + b)$. This is a B-spline with controls translated by b. The literal unsquared formula would not have this affine-invariance property: its translated-control expression adds $[1 + \frac{1}{2}t^2(1-t)]b$, not b.

Exercise 8.6.3: First and second derivatives of a cubic Bézier curve

(a) Differentiate the polynomial weights and regroup by differences.

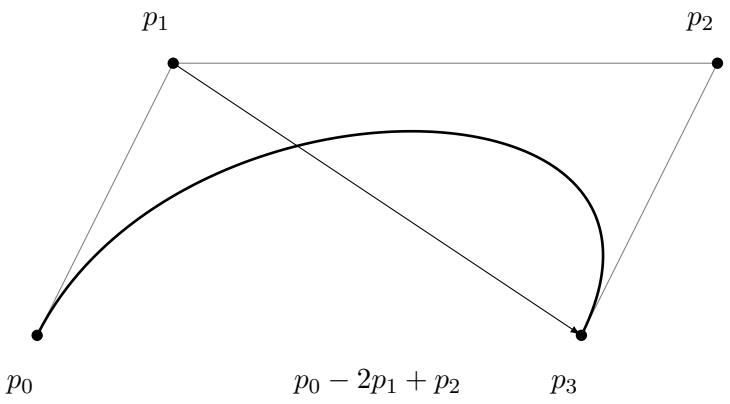
$$B'(t) = 3\big[(1-t)^2(p_1 - p_0) + 2t(1-t)(p_2 - p_1) + t^2(p_3 - p_2)\big].$$

In particular $B'(0) = 3(p_1 - p_0)$ and $B'(1) = 3(p_3 - p_2)$. The initial tangent direction goes from p_0 toward p_1; the final one goes from p_2 toward p_3, provided those differences are nonzero. Yes, $B'(1) = 0$ is possible: it occurs exactly when $p_3 = p_2$. At such an endpoint the zero tangent vector does not specify a direction.

(b) Differentiate once more.

$$B''(t) = 6\big[(1-t)(p_0 - 2p_1 + p_2) + t(p_1 - 2p_2 + p_3)\big].$$

Thus $B''(0) = 6(p_0 - 2p_1 + p_2)$ and $B''(1) = 6(p_1 - 2p_2 + p_3)$. With p_1 as origin, let $a = p_0 - p_1$, $c = p_2 - p_1$. Their vector sum $a + c = p_0 - 2p_1 + p_2$ has the direction of $B''(0)$. Construct the parallelogram on these two vectors; its diagonal gives the requested direction (unless the vector is zero). The schematic plot below uses controls $(0,0), (1,2), (5,2), (4,0)$ and shows that diagonal starting at p_1.



Exercise 8.6.4: Derivatives of the consistent cubic B-spline

(a) Compute the tangent and its endpoint values.

Differentiating the corrected formula gives

$$S'(t) = \tfrac{1}{2}\big[-(1-t)^2 p_0 + (3t^2 - 4t)p_1 + (-3t^2 + 2t + 1)p_2 + t^2 p_3\big].$$

Hence $S'(0) = (p_2 - p_0)/2$ and $S'(1) = (p_3 - p_1)/2$. The initial tangent is parallel to the segment from p_0 to p_2, and the final tangent to the segment from p_1 to p_3, when nonzero. If both vanish, then $p_2 = p_0 = a$ and $p_3 = p_1 = b$. Substitution gives

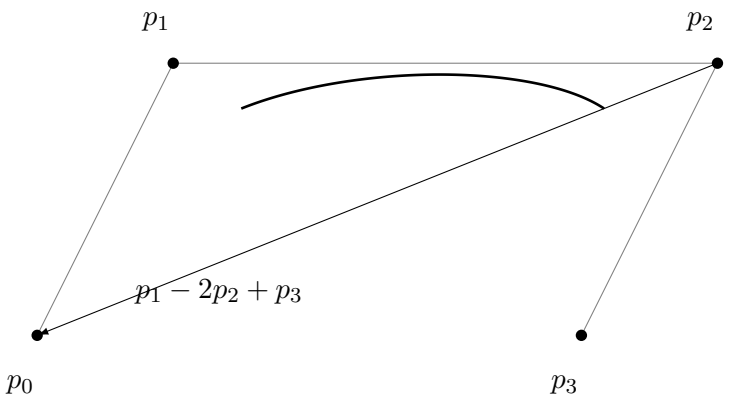
$$S(t) = (\tfrac{1}{3} + t^2 - \tfrac{2}{3}t^3)a + (\tfrac{2}{3} - t^2 + \tfrac{2}{3}t^3)b, \quad S'(t) = 2t(1-t)(a-b).$$

If $a \neq b$, this traverses the straight segment from $(a + 2b)/3$ to $(2a + b)/3$, with zero speeds at both endpoints but positive speed inside. It need not be constant. It is constant only if $a = b$.

(b) Compute the second derivative and its geometry.

$$S''(t) = (1-t)(p_0 - 2p_1 + p_2) + t(p_1 - 2p_2 + p_3).$$

In particular $S''(0) = p_0 - 2p_1 + p_2$ and $S''(1) = p_1 - 2p_2 + p_3$. Taking p_2 as origin, add the vectors $p_1 - p_2$ and $p_3 - p_2$; their parallelogram diagonal has the direction of $S''(1)$. The plot uses the same schematic controls as Exercise 8.6.3 but the B-spline curve and this second diagonal.



Exercise 8.6.5: Joining two cubic Bézier segments with matching velocities

(a) Equal derivatives give the midpoint condition.

At their shared control p_3, the first segment has derivative $3(p_3 - p_2)$ and the second has derivative $3(p_4 - p_3)$. Using the same unit parameter length for both pieces, C^1 continuity therefore requires and is guaranteed by

$$p_3 - p_2 = p_4 - p_3, \quad \text{or} \quad p_4 = 2p_3 - p_2.$$

Thus p_3 is the midpoint of p_2p_4.

(b) The zero-derivative case.

Both derivatives vanish exactly when $p_2 = p_3 = p_4$. The joined parametrization is still C^1, because its one-sided derivatives agree and are zero. But a zero velocity supplies no regular tangent direction; one cannot infer a smooth nonstationary join merely from parallel tangents. The remaining controls can give different departing and arriving directions at higher order, so C^2 continuity is not automatic.

Exercise 8.6.6: Adjacent B-spline segments automatically join

The first segment has controls p_0, p_1, p_2, p_3; the next has p_1, p_2, p_3, p_4.

(a) Check the endpoints.

By the endpoint formulas, $x(1) = (p_1 + 4p_2 + p_3)/6 = y(0)$, proving G^0 continuity.

(b) Check the derivatives with their actual scale factors.

The first segment has $x'(1) = (p_3 - p_1)/2$; the next has $y'(0) = (p_3 - p_1)/2$. They agree, so the combined unit-parameter curve is C^1 at the join. The last new control p_4 does not enter these equalities.

Exercise 8.6.7: A second derivative condition determines the next control

The C^1 condition from Exercise 8.6.5 gives $p_4 = 2p_3 - p_2$. The C^2 condition equates

$$x''(1) = 6(p_3 - 2p_2 + p_1), \qquad y''(0) = 6(p_3 - 2p_4 + p_5).$$

Cancel 6 and p_3, obtaining $p_5 = p_1 - 2p_2 + 2p_4$. Substitute the already determined p_4 to get

$$p_5 = p_1 - 4p_2 + 4p_3.$$

Thus p_4 and p_5 are fixed by the original controls and C^2 continuity; only the new segment's final control p_6 is free.

Exercise 8.6.8: Adjacent B-spline segments also match accelerations

For controls p_0, p_1, p_2, p_3, the final second derivative is $x''(1) = p_1 - 2p_2 + p_3$. For the next segment, with controls shifted forward by one, the initial second derivative is $y''(0) = p_1 - 2p_2 + p_3$. They are equal. Together with the endpoint and first-derivative equalities proved in Exercise 8.6.6, this establishes C^2 continuity at the join.

Exercise 8.6.9: The quartic Bézier basis matrix

Let $G = [\,p_0\ p_1\ p_2\ p_3\ p_4\,]$ and $u(t) = (1, t, t^2, t^3, t^4)^T$. Expand each of the five quartic weights in powers of t and put its coefficients in a *row* of M_B:

$$M_B = \begin{pmatrix} 1 & -4 & 6 & -4 & 1 \\ 0 & 4 & -12 & 12 & -4 \\ 0 & 0 & 6 & -12 & 6 \\ 0 & 0 & 0 & 4 & -4 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

For example its second row produces $4t - 12t^2 + 12t^3 - 4t^4 = 4t(1-t)^3$, the weight of p_1. Thus $x(t) = GM_Bu(t)$ reproduces the stated quartic curve.

Exercise 8.6.10: The uniform cubic B-spline basis matrix

Using $G = [\,p_0\ p_1\ p_2\ p_3\,]$ and $u(t) = (1, t, t^2, t^3)^T$, read the four consistent B-spline weights into rows:

$$M_S = \tfrac{1}{6}\begin{pmatrix} 1 & -3 & 3 & -1 \\ 4 & 0 & -6 & 3 \\ 1 & 3 & 3 & -3 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then $GM_Su(t)$ gives the B-spline formula from the Practice Problems. The factor $1/6$ is essential for normalization and cannot be dropped. The printed missing-square error in Exercise 2 is not used in this matrix.

Exercise 8.6.11: Three statements about Bézier curves

(a) **True.** A cubic Bézier curve uses four controls.

(b) **False.** For a quadratic curve the initial derivative is $2(p_1 - p_0)$, not $p_1 - p_0$. The direction is the same when nonzero, but the tangent vector has the extra factor two.

(c) **True.** The endpoint derivatives are $2(p_2 - p_1)$ and $2(p_3 - p_2)$, which agree exactly when $p_2 = (p_1 + p_3)/2$. The two pieces already share their endpoint, so this ensures C^1 continuity.

Exercise 8.6.12: Three statements about invariance and matrices

(a) **False.** Translations preserve the Bézier construction too, because the weights sum to one, as Exercise 8.6.1 proves.

(b) **True.** Equality of the ending and starting points is exactly G^0 continuity; matching derivatives requires an additional condition.

(c) **False.** The control columns form the geometry matrix G. The basis matrix contains fixed polynomial coefficients, independent of the controls.

Exercise 8.6.13: The first half's subdivided control points

(a) Use the endpoint derivative equation.

Equation (12) says $3(q_1 - q_0) = \frac{3}{2}(p_1 - p_0)$. Since $q_0 = p_0$, divide by 3 and rearrange: $q_1 = (p_0 + p_1)/2$.

(b) Rearrange the midpoint derivative equation.

Equation (13) gives $q_3 - q_2 = (-p_0 - p_1 + p_2 + p_3)/8$. Multiplying by eight and solving for q_2 gives $8q_2 = 8q_3 + p_0 + p_1 - p_2 - p_3$.

(c) Substitute the actual curve midpoint.

Use $q_3 = (p_0 + 3p_1 + 3p_2 + p_3)/8$ from equation (8). Then $8q_2 = 2p_0 + 4p_1 + 2p_2$, so

$$q_2 = (p_0 + 2p_1 + p_2)/4 = \tfrac{1}{2}[q_1 + (p_1 + p_2)/2].$$

Thus it is exactly the requested midpoint of two already constructed midpoints.

Exercise 8.6.14: The second half's subdivided controls

(a) Justify the endpoint derivative chain.

The equality $3(r_3 - r_2) = z'(1)$ is the Bézier endpoint formula. The equality $z'(1) = \frac{1}{2}x'(1)$ is the chain rule for $z(t) = x(1/2 + t/2)$. Finally $\frac{1}{2}x'(1) = \frac{3}{2}(p_3 - p_2)$ is the original endpoint formula.

(b) Solve for r_2.

Since $r_3 = p_3$, the equalities give $r_2 = (p_2 + p_3)/2$.

(c) Justify the initial derivative chain.

The identity $3(r_1 - r_0) = z'(0)$ is the initial Bézier derivative formula; $z'(0) = \frac{1}{2}x'(1/2)$ follows from the same reparametrization and chain rule.

(d) Use the original midpoint derivative.

Since $x'(1/2) = \frac{3}{4}(-p_0 - p_1 + p_2 + p_3)$, $3(r_1 - r_0) = \frac{3}{8}(-p_0 - p_1 + p_2 + p_3)$. Divide by 3 and rearrange to obtain $8r_1 = -p_0 - p_1 + p_2 + p_3 + 8r_0$.

(e) Substitute $r_0 = x(1/2)$.

The midpoint formula yields $8r_1 = 2p_1 + 4p_2 + 2p_3$. Consequently

$$r_1 = (p_1 + 2p_2 + p_3)/4 = \tfrac{1}{2}\big[r_2 + (p_1 + p_2)/2\big].$$

Exercise 8.6.15: A subdivision algorithm using only midpoint operations

(a) Apply the chain rule at the common midpoint.

With $y(t) = x(t/2)$ and $z(t) = x(1/2 + t/2)$, $y'(1) = \frac{1}{2}x'(1/2) = z'(0)$.

(b) Translate that equality into a control-point midpoint.

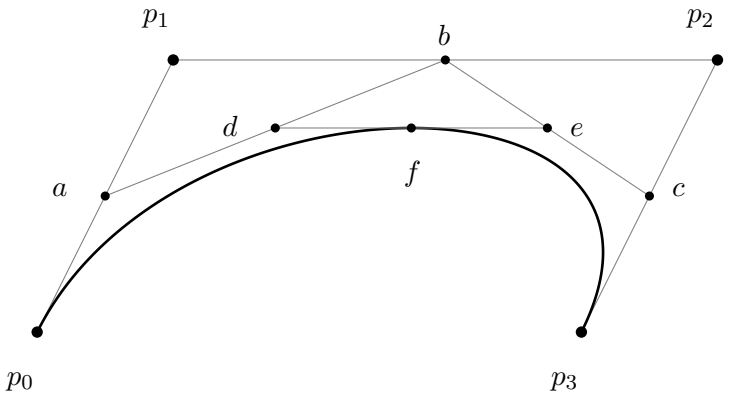
The endpoint formulas give $3(q_3 - q_2) = 3(r_1 - r_0)$. Using $q_3 = r_0$ and rearranging gives $2q_3 = q_2 + r_1$, so q_3 is the midpoint of $q_2 r_1$.

(c) State the full reusable algorithm.

Compute three first-level midpoints

$$a = (p_0 + p_1)/2, \quad b = (p_1 + p_2)/2, \quad c = (p_2 + p_3)/2;$$

then two second-level midpoints $d = (a + b)/2$, $e = (b + c)/2$; then the last midpoint $f = (d + e)/2$. The left controls are (p_0, a, d, f) and the right controls are (f, e, c, p_3). The formulas in Exercises 8.6.13–14 show $a = q_1, d = q_2, f = q_3 = r_0$, $e = r_1, c = r_2$. Thus the algorithm computes all required controls once, shares the repeated midpoint, and uses only sums and division by two.



Exercise 8.6.16: Endpoint positions and velocities determine the curve

The endpoint formulas recover all controls from the four specified vectors:

$$p_0 = x(0), \quad p_3 = x(1), \quad p_1 = x(0) + x'(0)/3, \quad p_2 = x(1) - x'(1)/3.$$

These equations determine one and only one control list, and substituting that list into the cubic Bézier formula determines one and only one curve. The derivative scale factor 3 is needed; just choosing any control points on the tangent lines would not determine the parameterized curve uniquely.

Exercise 8.6.17: Elevate a quadratic curve to cubic degree without changing it

(a) Match both endpoint positions and both endpoint velocities.

For $w(t) = (1-t)^2p_0+2t(1-t)p_1+t^2p_2$, $w(0) = p_0$, $w(1) = p_2$, $w'(0) = 2(p_1-p_0)$, and $w'(1) = 2(p_2-p_1)$. The preceding recovery formulas give cubic controls

$$r_0 = p_0, \quad r_1 = \tfrac{1}{3}p_0 + \tfrac{2}{3}p_1, \quad r_2 = \tfrac{2}{3}p_1 + \tfrac{1}{3}p_2, \quad r_3 = p_2.$$

Their endpoint derivatives are $3(r_1 - r_0) = w'(0)$ and $3(r_3 - r_2) = w'(1)$, so all four requirements are met.

(b) Check equality for every parameter.

Insert these controls into the cubic formula and collect the original p_0, p_1, p_2 coefficients. They become respectively

$$(1-t)^3 + t(1-t)^2 = (1-t)^2,$$
$$2t(1-t)^2 + 2t^2(1-t) = 2t(1-t), \qquad t^2(1-t) + t^3 = t^2.$$

Thus $x(t) = w(t)$ identically, including at both endpoints. The curve's polynomial degree representation has changed, but its geometry and parametrization have not.

Exercise 8.6.18: Multiply the scalar blocks by the control vectors

Treat the displayed p_j as vector blocks. Row-by-row block multiplication gives

$$\begin{pmatrix} p_0 \\ -3p_0 + 3p_1 \\ 3p_0 - 6p_1 + 3p_2 \\ -p_0 + 3p_1 - 3p_2 + p_3 \end{pmatrix}.$$

For controls in $\mathbb{R}^n$, this is a column of four vector blocks, or $4n$ scalar entries when expanded. These are the constant, linear, quadratic, and cubic vector coefficients of the curve; they agree with direct expansion of its four Bernstein weights.